

Three Duals at Surreal Scales

Strong Hahn Operators, Completion Defects,
and Invisible Functionals

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Abstract

Let $K = k((t^\Gamma))$ and $V_\Gamma = V((t^\Gamma))$, where V is a k -vector space and Γ is a set-sized ordered abelian group. We separate the algebraic scalar extension, its valuation completion, and its closure under strong Hahn summation. The completion consists exactly of those series whose coefficients below each valuation cut span a finite-dimensional k -space. In infinite coefficient dimension it equals the full Hahn space exactly when the nonzero group Γ is ordered isomorphic to $\mathbb{Z}$, and it is strictly larger than the algebraic scalar extension exactly when Γ has countable cofinality. Under a cofinal exponent enlargement this completion criterion is preserved; under a noncofinal enlargement its intersection with the original Hahn space collapses to the algebraic scalar extension.

We also classify every strong K -linear operator as a single Hahn series of arbitrary k -linear coefficient maps. The continuous dual has an exact restriction sequence: its image consists of pointwise Hahn-valued coefficient functionals with a common lower support bound, while its kernel consists of continuous functionals vanishing on every constant vector. These latter functionals separate precisely the vectors outside the completed scalar extension. For every infinite-dimensional V and noncyclic $\Gamma \neq 0$, there are two distinct failures of strongness: continuous maps can destroy summability, or preserve the summability of a particular monomial family but fail to commute with its sum.

For infinite-dimensional complex Hilbert H and noncyclic Γ , this produces three strictly nested duals: inner-product-represented, strong, and valuation-continuous. The first gap is exactly the Hahn extension of the ordinary algebraic-dual quotient $H^\# / H'$. All strong maps are bounded by a positive real Hahn scalar, even when their coefficient maps are discontinuous in the ordinary Hilbert norm. An explicit choice-based hyperplane has a strong bounded projection but zero orthogonal complement, and the distance from an exterior constant vector has no infimum in the scalar field. Section 12 compares these results with the collection's *Infinite-Dimensional Hahn Spectral Theory*, whose Hahn-Hilbert space is the one used here.

Scope and status. Complete mathematical proofs are supplied for the stated results. The proposed contribution is the combined completion–duality–scalar-enlargement classification, not the invention of Hahn summation, non-Archimedean Hahn–Banach, or failures of Riesz representation. No named published conjecture is claimed solved. A targeted repository and primary-literature comparison did not locate this exact package; this is not a certification of priority. This is an AI-assisted research draft: the proofs have not been independently refereed or formally verified in Lean or any other proof assistant. All workspaces and choices in the proofs are set-sized.

Provenance. Built from one manuscript, delivered on September 22, 2026 as item 04 of the collection's nineteenth batch and written against repository revision 048b72c; placed at 52c7ab6. Editorial changes, the updated repository comparison and the ledger of non-claims are in appendix A.

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1 The questions and the contribution

1.1 Why continuity is not the same as respecting a formal sum

An infinite-dimensional extension of complex analysis to surcomplex scalars has to specify more than its scalar field. It must specify its vectors, the admissible infinite sums, and the meaning of continuity. For the natural coefficient space

$$H((t^\Gamma)) = \left\{ \sum_{\gamma \in \Gamma}^H h_\gamma t^\gamma : h_\gamma \in H, \text{ supp } h \text{ is well ordered} \right\},$$

three plausible requirements on a scalar-valued linear functional are not interchangeable. It may be given by an inner product. It may commute with every admissible Hahn sum. Or it may merely be continuous when closeness means agreement below a sufficiently high valuation cut.

The issue is visible before introducing a Hilbert norm. Given an ordinary vector space V , form the algebraic scalar extension $K \otimes_k V$. It embeds in $V((t^\Gamma))$, and its vectors have coefficients in one finite-dimensional subspace of V . Completing this subspace need not give all of $V((t^\Gamma))$. A Hahn sum can contain infinitely many linearly independent coefficients below a single valuation cut. Such a vector is permitted by well-ordered support, but no finite coefficient span can match it to that precision.

This distinction leads to four concrete questions. Which Hahn vectors belong to the completed algebraic scalar extension? Which linear maps respect all strong sums? What can a continuous functional detect that its values on constants do not determine? What happens to these answers when additional surreal scales are admitted? The theorems below give exact answers, including the exceptional cases.

1.2 Main statements at a glance

For a nonzero ordered abelian group Γ , write $V_\Gamma = V((t^\Gamma))$, $K = k((t^\Gamma))$,

$$E_\Gamma(V) = \text{span}_K V, \quad C_\Gamma(V) = \overline{E_\Gamma(V)}^v.$$

Our convention is that a strong map both preserves admissibility of all strongly summable families and commutes with their sums.

Theorem 1.1 (Completion and enlargement; proved in sections 4 and 10). *For every $x \in V_\Gamma$,*

$$x \in E_\Gamma(V) \iff \dim_k \text{span}\{x_\delta : \delta \in \Gamma\} < \infty, \quad (1.1)$$

$$x \in C_\Gamma(V) \iff (\forall \gamma \in \Gamma) \dim_k \text{span}\{x_\delta : \delta < \gamma\} < \infty. \quad (1.2)$$

If V is infinite-dimensional, then $C_\Gamma(V) = V_\Gamma$ exactly when $\Gamma \simeq_{\text{ord}} \mathbb{Z}$. For an ordered subgroup inclusion $\Gamma \subseteq \Delta$,

$$C_\Delta(V) \cap V_\Gamma = \begin{cases} C_\Gamma(V), & \Gamma \text{ is cofinal in } \Delta, \\ E_\Gamma(V), & \Gamma \text{ is not cofinal in } \Delta. \end{cases}$$

Theorem 1.2 (Strong maps and the continuous dual; proved in sections 3 and 6). *For arbitrary k -vector spaces V, W ,*

$$\text{Hom}_K^{\text{str}}(V_\Gamma, W_\Gamma) \simeq \text{Hom}_k(V, W)((t^\Gamma)).$$

Let $P_\Gamma(V)$ be the space of k -linear maps $A : V \rightarrow K$ for which $v(Au) \geq s$ for all $u \in V$, with some common $s \in \Gamma$. There is an exact sequence

$$0 \longrightarrow N_\Gamma(V) \longrightarrow \text{Hom}_K^{\text{cont}}(V_\Gamma, K) \xrightarrow{\text{res}} P_\Gamma(V) \longrightarrow 0,$$

where $N_\Gamma(V)$ consists of the continuous functionals zero on V . Their common kernel is exactly $C_\Gamma(V)$.

For infinite-dimensional V and noncyclic Γ , the kernel $N_\Gamma(V)$ is nonzero. Moreover, some restrictions in $P_\Gamma(V)$ are not the restrictions of any strong map: their individual values are Hahn series, but their *global* coefficient support is not well ordered. Thus passing from strong maps to continuous maps adds two separate kinds of freedom, not just an undetectable additive error.

For a complex Hilbert coefficient space the additional comparison with inner products is

$$D_{\text{rep}} \subseteq D_{\text{str}} \subseteq D_{\text{cont}}, \quad D_{\text{str}}/D_{\text{rep}} \simeq (H^\# / H')((t^\Gamma)). \quad (1.3)$$

Both inclusions are strict when H is infinite-dimensional and Γ is noncyclic. For infinite-dimensional H and $\Gamma \simeq_{\text{ord}} \mathbb{Z}$, only the first is strict. In finite coefficient dimension all three coincide.

1.3 What has already been done, and what is not claimed

The repository's *Infinite-Dimensional Hahn Spectral Theory* already constructs the coefficient-convolution inner product and proves that every everywhere-defined adjointable operator is a Hahn series of ordinary bounded operators (`ihshh:thm:adjoint`). It also proves spectral and closed-range phenomena for that space [9], including a valuation-continuous functional that is not represented by an inner product (`ihshh:prop:riesz`) and a valuation-closed proper subspace with zero orthogonal complement (`ihshh:thm:closedrange`). We do not present those results as new; the last two are the nearest predecessors, inside the collection, of Theorems 8.4 and 9.1 below, and section 12 states exactly how they relate. Our operator theorem replaces the adjoint hypothesis by preservation of strong sums, permits *arbitrary algebraic* coefficient maps, and is then used to compare with the continuous dual, which is strictly larger in infinite coefficient dimension for noncyclic Γ . The coefficient-rank completion criterion and its cofinal/noncofinal enlargement dichotomy address a different question from the repository's persistence of spectral range defects.

Formal summability spaces, strongly linear maps, and the resulting operator algebras are established notions. In particular, Bagayoko–Krapp–Kuhlmann–Panazzolo–Serra give a general axiomatic framework and discuss nonstrong linear maps, referring to earlier examples [1, Section 1.3]. Our definition follows that usage. Their scalar-field linearity conventions must not be conflated with ours: the main operator classification here is *linear over the entire Hahn field K* , not merely over its coefficient field k . In particular, our Hahn summability is stable under multiplication of the entire family by one fixed element of K , but not under arbitrary term-dependent K -scalars. For $u \neq 0$ and $\eta > 0$, the family $(t^{\eta n}u)_{n \geq 1}$ is strong, whereas multiplying its n th term by $t^{-\eta n}$ gives the nonsummable constant family $(u)_{n \geq 1}$. Thus the extra ultrafiniteness axiom of [1, Definition 1.1] holds here over k : coefficient scalars introduce no new exponents, and each exponent has finitely many original contributors, each replaced by finitely many scalar multiples. It does not hold in general over K .

The extension argument belongs to the classical non-Archimedean Hahn–Banach tradition initiated by Ingleton [2]; Morillon states its one-step form for real-valued ultrametric seminorms over spherically complete valued fields and discusses its choice requirements [3, Section 3.1.2, Lemma 2].

That formulation does not itself cover arbitrary ordered value groups; we give the needed ordered-group ball argument and Zorn extension in full. Spherical completeness and extension are tools, not claimed innovations. Non-Archimedean failures of orthogonal complementation are likewise established; Aguayo–Nova study them in a c_0 -based model [4]. Our final hyperplane theorem specifies a different coefficient-Hilbert model and an exact missing-infimum cut.

The proposed originality lies in the precise classification package, especially Theorems 4.2, 4.5, 6.3, 6.4, 7.1 and 10.1. The inspected repository inventory and the primary sources consulted are recorded in section 14 and the accompanying audit file. This comparison is deliberately narrower than an assertion that no equivalent theorem exists anywhere in print.

1.4 Conventions, and words used elsewhere in the collection

Several words below are also used by neighbouring reports, sometimes for other things. Each has exactly one meaning in this report.

- *Strong* (map, functional, dual) means preserving strong Hahn summability of every set-indexed family *and* commuting with its sum (Definition 2.1 and the definition in section 3). This is the “strongly linear” of the dynamics report (`dyn:def:summable`) and of [1], here required to be linear over the whole Hahn field K . It has nothing to do with the classical strong operator topology, a distinction the collection’s notation guide also records.
- *Continuous* and *completion* refer to the intrinsic valuation topology and uniformity of the one fixed workspace $V((t^\Gamma))$, with thresholds in Γ . They never refer to the full surreal fine topology, in which a set-sized workspace is discrete (`ih:s:warn:three`), nor to the ordinary norm topology of a coefficient space. The completion $C_\Gamma(V)$ is a closure inside V_Γ ; it is not a sequence-space completion such as $\ell^2(K)$.
- *Dual* always means a space of K -linear or k -linear functionals: $V^\#$ is the ordinary algebraic dual, H' the ordinary continuous dual, and $D_{\text{rep}} \subseteq D_{\text{str}} \subseteq D_{\text{cont}}$ the three duals of section 8. It never means an order dual of an exponent group.
- *Cofinality* $\text{cf}(\Gamma)$ is the least cardinality of a cofinal subset of the ordered set Γ , and “ Γ is cofinal in Δ ” has its order-theoretic meaning; both agree with the report *Cofinality, Factorization and Scalar Extension for Entire Hahn Functions at Arbitrary Rank* [10] (`ent:rem:one-name`, `ent:thm:extension-criterion`). An element $\alpha > 0$ whose multiples $n\alpha$ are cofinal is called an *order unit* there (`ent:def:order-unit`); countable cofinality does not imply that an order unit exists.
- *Noncyclic* is used only for $\Gamma \neq 0$ and means that Γ is not ordered isomorphic to $\mathbb{Z}$. For a nonzero ordered group this is the same as not being cyclic.
- *Bounded* for a map on a Hilbert coefficient space means an inequality with a positive bound in $F = \mathbb{R}((t^\Gamma))$ (Proposition 8.2); ordinary boundedness of coefficient maps is always called ordinary boundedness or ordinary continuity.
- *Invisible* is always “invisible on constants” (section 6). It is unrelated to the dynamics report’s remark on divisor growth that the valuation does not see (`dyn:rem:invisible`).
- Inner products are linear in the first variable, as in Part I of the spectral report and unlike its Part II. We write $H_\Gamma = H((t^\Gamma))$; this is the space that report calls $\mathcal{H}_\Gamma$. The letters E and C of $E_\Gamma(V)$ and $C_\Gamma(V)$ are unrelated to the perturbation E and the coercive operator C used there.

2 Hahn vector spaces and the two notions of infinity

2.1 Supports, scalar multiplication, and strong families

Fix a field k and a set-sized ordered abelian group Γ , written additively. Its order is translation invariant. For a k -vector space V , a Hahn vector is a function $x : \Gamma \rightarrow V$ whose nonzero support is well ordered in the given order. We use formal notation

$$x = \sum_{\gamma \in \Gamma}^H x_\gamma t^\gamma, \quad V_\Gamma = V((t^\Gamma)).$$

The zero vector has empty support. The scalar field is $K = k((t^\Gamma))$. Its multiplication and its action on V_Γ are coefficient convolutions. In particular,

$$(ax)_\eta = \sum_{\alpha+\beta=\eta} a_\alpha x_\beta.$$

The superscript H emphasizes formal Hahn summation; it never means ordinary summation of infinitely many coefficients at one exponent.

Definition 2.1. A set-indexed family $(x_j)_{j \in J}$ in V_Γ is strongly Hahn summable if $\bigcup_j \text{supp } x_j$ is well ordered and, for every γ , only finitely many j have $(x_j)_\gamma \neq 0$. Its sum is defined by the finite coefficient sums $[t^\gamma](\sum_j^H x_j) = \sum_j (x_j)_\gamma$.

Every Hahn vector is the strong sum of its monomial coefficient vectors. An infinite family of nonzero constants is not strong. Even an ordinary convergent numerical series, such as $\sum_{n \geq 1} 2^{-n}$, is not that family's Hahn sum: the exponent zero has infinitely many contributors. The familiar numerical value may be used as a *single coefficient*, which is a different operation.

Lemma 2.2 (Classical support facts). *If $A, B \subseteq \Gamma$ are well ordered, then $A + B$ is well ordered, and each η has only finitely many representations $\eta = \alpha + \beta$ with $\alpha \in A$, $\beta \in B$. A finite union of well-ordered subsets is well ordered. If $S \subseteq \Gamma_{>0}$ is well ordered, the monoid of finite sums of elements of S is well ordered; for each exponent there are only finitely many words in S having that sum, allowing all finite lengths.*

Proof. We use choice, as throughout the report, so a linearly ordered set is well ordered exactly when it has no infinite strictly decreasing sequence. Every sequence in a well-ordered set has an infinite nondecreasing subsequence: choose an index attaining the minimum of the current tail, then repeat strictly after that index. Suppose $\alpha_n + \beta_n$ were strictly decreasing, with $\alpha_n \in A$ and $\beta_n \in B$. On a subsequence with nondecreasing α_n , the β_n must strictly decrease, contrary to well-ordering of B . Hence $A + B$ is well ordered. If one η had infinitely many distinct representations, select a sequence of them. Their A -coordinates are distinct, since the B -coordinate equals $\eta - \alpha$. A nondecreasing subsequence is therefore strictly increasing in A , forcing a strictly decreasing sequence in B . This proves finiteness of each coefficient sum. Empty supports cause no exception. For a finite union, intersect a nonempty subset with each member of the union and take the least of the finitely many available minima. The positive-monoid assertion is the classical Hahn–Neumann support lemma [5], stated in the collection as `ihf:lem:support` and proved in the finite-dimensional spectral report as `spec:lem:words` and `spec:lem:neumann`; it is needed below only for the standard binomial square-root construction. The operator and completion classifications use the two-set assertion, whose proof was just given. $\square$

Strong sums may be regrouped along any map $f : J \rightarrow I$ of index sets. Each fiber subfamily is strong; give an empty fiber sum the value zero. At each exponent only finitely many original indices contribute, hence only finitely many fibers can contribute. The union of the regrouped supports is a subset of the original well-ordered union. Thus the fiber sums form a strong family and their sum is the original sum, by finite coefficient addition.

Scalar multiplication also commutes with strong sums. If A is the scalar support and $B = \bigcup_j \text{supp } x_j$, all product supports lie in $A + B$. For a fixed output exponent there are finitely many pairs from $A \times B$; for each second coordinate only finitely many j occur. This permits the coefficient rearrangement without any infinite sum in V . Applying the two-set support lemma twice similarly gives finiteness for three convolution factors, proving associativity of the scalar action. The identity scalar acts through its sole coefficient at zero. These facts establish the K -vector-space structure and supply the infinite rearrangements used below.

2.2 Valuation topology and bounded shifts

Define

$$v(x) = \min \text{supp } x \quad (x \neq 0), \quad v(0) = +\infty.$$

For $a \in K^\times$, $v(ax) = v(a) + v(x)$, and $v(x + y) \geq \min\{v(x), v(y)\}$. The equality for scalar multiplication holds because the product's first coefficient is a nonzero scalar multiple of the first vector coefficient. It entails no norm on V .

Except where expressly stated, assume $\Gamma \neq 0$. The intrinsic valuation topology has a neighborhood basis

$$U_\gamma(x) = \{y : v(y - x) > \gamma\}, \quad \gamma \in \Gamma.$$

Closed valuation balls $B(x, \gamma) = \{y : v(y - x) \geq \gamma\}$ give equivalent precision tests: one can replace γ by $\gamma + \eta$, with any fixed $\eta > 0$. The topology is Hausdorff, since a nonzero difference has a finite valuation and can be excluded by a higher precision.

Lemma 2.3 (Continuity is a uniform valuation-shift bound). *A K -linear map $L : V_\Gamma \rightarrow W_\Gamma$ is valuation-continuous if and only if there exists $s \in \Gamma$ such that*

$$v(Lx) \geq v(x) + s \quad (x \in V_\Gamma). \quad (2.1)$$

The same assertion holds for a K -linear map on any subspace with the induced valuation topology.

Proof. Suppose the bound holds. For a target neighborhood $U_\gamma(0)$, the condition $v(x) > \gamma - s$ implies $v(Lx) > \gamma$; this proves continuity at zero, and additivity proves continuity everywhere. Conversely, continuity at zero for the target neighborhood $U_0(0)$ supplies $\delta \in \Gamma$ such that $v(z) > \delta$ implies $v(Lz) > 0$. Fix $\eta > 0$. For nonzero x , put $a = t^{\delta + \eta - v(x)}$. Then $v(ax) = \delta + \eta$, so

$$v(Lx) > v(x) - \delta - \eta.$$

Here $L(ax) = aLx$ and the scalar valuation identity justify subtracting $v(a)$; if $Lx = 0$ the desired inequality is immediate. Thus $s = -\delta - \eta$ gives the required weak inequality. The case $x = 0$ is also immediate, with the convention $+\infty + s = +\infty$. The argument uses only scalar stability of the domain, so it works on any subspace with its induced topology, including the zero subspace. $\square$

For $\Gamma = 0$, both Hahn spaces reduce to their coefficient spaces with the discrete valuation topology: $U_0(x) = \{x\}$. Every linear map is continuous and satisfies the shift inequality with $s = 0$. Later statements expressly including the zero group use this direct argument, without choosing a positive η .

Warning 2.4 (Constants carry the discrete valuation topology). If V is originally normed, its embedded constant copy in V_Γ does not carry that norm topology: every nonzero constant has valuation zero. A map discontinuous in the ordinary norm can therefore become continuous after coefficientwise Hahn extension. This is not a contradiction to ordinary functional analysis.

3 All strong linear maps have one global operator support

3.1 The exact Hom theorem

Definition 3.1. A K -linear map $T : V_\Gamma \rightarrow W_\Gamma$ is strong if, for every strongly Hahn-summable family (x_j) , the family (Tx_j) is strongly Hahn summable and $T(\sum_j^H x_j) = \sum_j^H Tx_j$.

Both clauses are part of the definition. In section 7 continuous functionals will exhibit two different obstructions.

Theorem 3.2 (Global-support classification). *For all k -vector spaces V, W and every ordered abelian group Γ , including the zero group, the convolution map gives a canonical K -linear isomorphism*

$$\mathrm{Hom}_k(V, W)((t^\Gamma)) \xrightarrow{\sim} \mathrm{Hom}_K^{\mathrm{str}}(V_\Gamma, W_\Gamma). \quad (3.1)$$

Explicitly, a Hahn series $A = \sum_\gamma^H A_\gamma t^\gamma$ acts by

$$(Ax)_\eta = \sum_{\gamma+\delta=\eta} A_\gamma(x_\delta). \quad (3.2)$$

The set $\{\gamma : A_\gamma \neq 0\}$ is one well-ordered support, independent of x . No topology on V or W is assumed.

Proof. Necessity of one global support. Let T be strong. For a constant $u \in V$ define $A_\gamma(u) = [t^\gamma]Tu$. Coefficient extraction shows that every A_γ is k -linear. Put $S = \{\gamma : A_\gamma \neq 0\}$. If S were not well ordered, choice would provide a strictly decreasing sequence $\gamma_1 > \gamma_2 > \dots$ in S . Select $u_n \in V$ with $A_{\gamma_n}(u_n) \neq 0$.

The family $x_n = t^{-\gamma_n}u_n$ is strong: its singleton exponents form a strictly increasing sequence, and every exponent occurs only once. But K -linearity gives $Tx_n = t^{-\gamma_n}Tu_n$, whose coefficient at exponent zero is $A_{\gamma_n}(u_n) \neq 0$ for every n . The input support is well ordered because it is the range of a strictly increasing sequence, even if that sequence is bounded above. Its images violate the finite-fiber condition at the single exponent zero. This contradicts strongness and proves that S is well ordered.

Determination by constants. Every x is the strong sum of $t^\delta x_\delta$ over its support. Strongness and K -linearity yield

$$Tx = \sum_\delta^H t^\delta T(x_\delta) = \sum_{\delta, \gamma}^H A_\gamma(x_\delta) t^{\delta+\gamma}.$$

The common support S and Lemma 2.2 justify the final expansion and make its coefficient formula exactly (3.2). Testing constants also proves uniqueness of the coefficient maps.

Sufficiency. Start instead with a well-ordered support of maps A_γ . For one input x , Lemma 2.2 makes (3.2) a Hahn vector, since its support is contained in $S + \mathrm{supp} x$. Additivity follows coefficientwise. For $a \in K$, both $A(ax)$ and $a(Ax)$ expand at exponent η over triples (γ, α, δ) with

$\gamma + \alpha + \delta = \eta$ in the three fixed supports. The two-set lemma applied twice makes this set finite, and k -linearity gives $A_\gamma(a_\alpha x_\delta) = a_\alpha A_\gamma(x_\delta)$. Thus the operator is K -linear.

If (x_j) is strong, let $B = \bigcup_j \text{supp } x_j$. All image supports lie in $S + B$. At a fixed exponent there are finitely many pairs (γ, δ) , and for each such δ finitely many j contribute. Thus the image family is strong, and the same finite rearrangement proves commutation with its sum.

Finally the correspondence is additive, and multiplication of A by a Hahn scalar corresponds to multiplication of the resulting operator by that scalar, using the same finite triple sums. Together with determination by constants, this proves the claimed canonical K -linear isomorphism. If $\Gamma = 0$, strong families have only finitely many nonzero terms, and the theorem reduces to $\text{Hom}_k(V, W) = \text{Hom}_k(V, W)$. $\square$

The descending-support test is an automatic uniformity principle: an assumption about all summable families rules out a global support defect that may not be visible on any one constant vector. It does not use Baire category, completeness of V , or an adjoint. By contrast, the automatic adjoint theorem of the spectral report obtains its one global support by a Baire argument on bounded coefficient operators (`ihshh:thm:adjoint`, Step 2); section 12 compares the two operator classes.

Corollary 3.3 (Operator calculus and automatic continuity). *Composition corresponds to coefficient convolution. In particular,*

$$\text{End}_K^{\text{str}}(V_\Gamma) \simeq \text{End}_k(V)((t^\Gamma))$$

as unital algebras. If $A \neq 0$ and $s = \min \text{supp } A$, then $v(Ax) \geq v(x) + s$ for all x . Every strong map is valuation-continuous.

Proof. For composable A, B , the coefficient of $B \circ A$ at η is $\sum_{\beta+\alpha=\eta} B_\beta A_\alpha$. All sums are finite by Lemma 2.2, and the action on a vector agrees after regrouping. The identity has the sole potentially nonzero operator coefficient id_V at exponent zero. The scalar coefficients act centrally, so the correspondence is one of unital K -algebras, even though operator composition need not commute. For the estimate, when $x \neq 0$ every nonzero output coefficient lies at a sum of an exponent at least s and one at least $v(x)$. Cancellation can raise the valuation, so the claim is an inequality. The cases $x = 0$ or $A = 0$ are immediate. For $\Gamma \neq 0$ apply Lemma 2.3; for $\Gamma = 0$ use the discrete topology described above. $\square$

3.2 Pointwise supports alone do not suffice

For a mere k -linear map $A : V \rightarrow K$, every individual $A(u)$ has a well-ordered support. It need not follow that

$$\bigcup_{u \in V} \text{supp } A(u)$$

is well ordered. For a concrete example, let V have a Hamel basis containing distinct e_n , $n \geq 1$, take $\Gamma = \mathbb{Q}$ and $\delta_n = 1/n$, and set $A(e_n) = t^{\delta_n}$, with zero values on all remaining basis vectors. Each vector is a finite basis combination, so every $A(u)$ has finite support, contained in $\{\delta_n : n \geq 1\}$. The global support is that entire strictly decreasing sequence. The vector

$$x = \sum_{n \geq 1}^H t^{-\delta_n} e_n$$

is a valid Hahn vector, but attempting the coefficient formula for its image puts one contribution 1 at exponent zero for every n . This is not a finite coefficient sum, over any characteristic of k . Thus no strong extension can have these values on constants.

On the other hand $v(Au) \geq 0$ for every u , so this restriction does satisfy a common lower valuation bound. It has a continuous, noncanonical extension by Theorem 6.3 below. The same example works whenever Γ admits a decreasing positive sequence; every noncyclic nonzero group does, by subtracting the bounded increasing sequence of Lemma 4.4 from a strict upper bound. The assumption of an infinite independent family in V is essential to this construction.

4 The completion is a coefficient-rank condition

4.1 The algebraic scalar extension inside the Hahn space

The constant embedding $V \hookrightarrow V_\Gamma$ induces a map $K \otimes_k V \rightarrow V_\Gamma$. If $u_1, \dots, u_d$ are linearly independent in V and $a_1, \dots, a_d \in K$, then

$$\sum_{j=1}^d a_j u_j = 0 \implies \sum_{j=1}^d [t^\delta] a_j u_j = 0 \quad \text{for every } \delta.$$

Independence makes every scalar coefficient zero. Thus the tensor map is injective and its image is $E_\Gamma(V) = \text{span}_K V$.

Proposition 4.1 (Finite total coefficient rank). *A vector $x \in V_\Gamma$ belongs to $E_\Gamma(V)$ if and only if all its coefficients span a finite-dimensional k -subspace of V .*

Proof. The coefficients of a finite sum $\sum a_j u_j$ lie in the span of its finitely many u_j . Conversely, let $u_1, \dots, u_d$ be a basis of the coefficient span and write $x_\delta = \sum_j c_{j\delta} u_j$. Each scalar series $a_j = \sum_\delta^H c_{j\delta} t^\delta$ has support contained in $\text{supp } x$, hence is a Hahn scalar. Then $x = \sum_{j=1}^d a_j u_j$. $\square$

Finite total coefficient rank is much weaker than finite support. A single constant vector multiplied by an arbitrarily long Hahn scalar series still lies in $E_\Gamma(V)$.

4.2 Exact characterization of the closure

For a cut level $\gamma \in \Gamma$ define

$$V_{x, < \gamma} = \text{span}_k \{x_\delta : \delta < \gamma\}.$$

The coefficients may have a transfinite well-ordered support, but this is an ordinary set-generated vector subspace.

Theorem 4.2 (Finite rank below every cut). *For $\Gamma \neq 0$,*

$$C_\Gamma(V) = \{x \in V_\Gamma : \dim_k V_{x, < \gamma} < \infty \text{ for every } \gamma \in \Gamma\}. \quad (4.1)$$

This is the completion of $K \otimes_k V$ for the valuation uniformity pulled back through its injective embedding in V_Γ . It is a K -subspace. Every vector of V_Γ is a strong sum of vectors of $E_\Gamma(V)$, whether or not it lies in $C_\Gamma(V)$.

Proof. Suppose x lies in the closure. For any γ , choose $y \in E_\Gamma(V)$ with $v(x - y) \geq \gamma$. Their coefficients below γ agree, and by Proposition 4.1 those of y have finite span. Hence $V_{x, < \gamma}$ is finite-dimensional.

Conversely, assume the right-hand condition. For any γ , the truncation

$$x_{< \gamma} = \sum_{\delta < \gamma}^H x_\delta t^\delta$$

has finite total coefficient rank. It belongs to $E_\Gamma(V)$ and satisfies $v(x - x_{< \gamma}) \geq \gamma$. For an open neighborhood $U_\rho(x)$ choose $\gamma > \rho$, possible because $\Gamma \neq 0$. Then $x_{< \gamma} \in U_\rho(x)$. In fact the truncations form an explicit convergent net, indexed by Γ directed by its order: for every ρ , all later cuts $\gamma > \rho$ have this property. There is no assumption of a countable cofinal set of cuts.

A closure of a vector subspace in a topological vector space is a subspace. Alternatively, addition combines two finite coefficient spans; the zero scalar is immediate, and if $a \neq 0$, every contribution $a_\alpha x_\beta$ to a coefficient of ax below γ satisfies $\alpha \geq v(a)$ and $\alpha + \beta < \gamma$. Thus $\beta < \gamma - v(a)$, and all these coefficients lie in the fixed finite-dimensional space $V_{x, < \gamma - v(a)}$. The ambient space is Hausdorff and complete by Proposition 5.1 below. Its closed subspace $C_\Gamma(V)$ is therefore complete, and $E_\Gamma(V)$ is dense in it by construction. With the stated induced uniformity, this identifies it as a completion of the embedded tensor product. Finally $x = \sum_\delta^H t^\delta x_\delta$ is always a strong sum of elements of $E_\Gamma(V)$. $\square$

Example 4.3 (A bounded increasing support). Let $e_2, e_3, \dots$ be linearly independent and let Γ be nonzero and divisible. Choose $\alpha > 0$ and put

$$x = \sum_{n \geq 2}^H t^{\alpha(1-1/n)} e_n. \quad (4.2)$$

Its support is a strictly increasing sequence and is well ordered. Every exponent is below α , while the coefficients below that cut have infinite rank. Consequently $x \notin C_\Gamma(V)$. All finite partial sums belong to $E_\Gamma(V)$, but none approximates x with error of valuation at least α . Nor can any other vector of finite total coefficient rank do so.

The final sentence is essential. Showing only that the displayed partial sums fail to converge would not exclude a different family of approximants. The rank criterion excludes *every* algebraic scalar-extension approximant.

4.3 Exactly which exponent groups make the algebraic part dense?

Lemma 4.4 (An order-theoretic alternative). *For a nonzero ordered abelian group Γ , precisely one of the following holds: $\Gamma \simeq_{\text{ord}} \mathbb{Z}$, or Γ contains a strictly increasing sequence bounded above in Γ . Equivalently, every well-ordered subset has finite intersection with every bounded initial interval exactly in the first case.*

Proof. If Γ has no least positive element, choose $\delta_0 > \delta_1 > \dots > 0$. The sequence $\delta_0 - \delta_n$ for $n \geq 1$ is strictly increasing and bounded above by δ_0 .

Suppose instead that ε is the least positive element. If $\Gamma = \mathbb{Z}\varepsilon$, the asserted cyclic description holds. Otherwise take a nonzero element outside $\mathbb{Z}\varepsilon$ and change its sign if necessary to obtain $\beta > 0$ outside that subgroup. It must be larger than every $n\varepsilon$. For if n were the least positive integer with $n\varepsilon > \beta$, then $0 < \beta - (n-1)\varepsilon < \varepsilon$, a contradiction. Equality $\beta = n\varepsilon$ is excluded. Thus $n\varepsilon$ is a bounded increasing sequence.

In an ordered copy of $\mathbb{Z}$, a nonempty well-ordered subset has a least integer and therefore finitely many elements below any fixed integer. Conversely, the increasing bounded sequence just constructed is itself a well-ordered subset violating that property. $\square$

Theorem 4.5 (The cyclic-group boundary). *If $\dim_k V < \infty$, then $E_\Gamma(V) = V_\Gamma$ for every Γ . If $\dim_k V = \infty$ and $\Gamma \neq 0$, then*

$$C_\Gamma(V) = V_\Gamma \iff \Gamma \simeq_{\text{ord}} \mathbb{Z}.$$

*For infinite-dimensional V and any nonzero Γ , the first inclusion $E_\Gamma(V) \subseteq C_\Gamma(V)$ is strict whenever there exists a sequence $n\alpha$ cofinal in Γ , with $\alpha > 0$, that is, whenever Γ has an order unit in the sense of **ent: def: order-unit**.*

Proof. Finite dimension follows from Proposition 4.1. In the cyclic case, every support has finitely many exponents below each cut; apply Theorem 4.2. In the noncyclic case take a bounded increasing sequence $\gamma_n < \beta$ from Lemma 4.4, choose linearly independent e_n , and form $\sum_n^H t^{\gamma_n} e_n$. It violates the finite-rank condition below β .

For the last assertion, $x = \sum_{n \geq 1}^H t^{n\alpha} e_n$ has infinite total coefficient rank but only finitely many support terms below each cut, by the stated cofinality. Hence it belongs to $C_\Gamma(V) \setminus E_\Gamma(V)$. $\square$

The last clause uses a sufficient condition that covers many familiar workspaces. There is also an exact criterion for whether completion adds *any* new vectors.

Theorem 4.6 (The cofinality boundary for nontrivial completion). *Suppose $\dim_k V = \infty$ and $\Gamma \neq 0$. Then*

$$E_\Gamma(V) \subsetneq C_\Gamma(V) \iff \text{cf}(\Gamma) = \aleph_0,$$

where $\text{cf}(\Gamma)$ is the least cardinality of a cofinal subset. Consequently the complete diagram is

<i>Exponent group</i>	<i>Algebraic, completed, and full Hahn spaces</i>
$\Gamma \simeq_{\text{ord}} \mathbb{Z}$	$E_\Gamma(V) \subsetneq C_\Gamma(V) = V_\Gamma$
Γ noncyclic, $\text{cf}(\Gamma) = \aleph_0$	$E_\Gamma(V) \subsetneq C_\Gamma(V) \subsetneq V_\Gamma$
$\text{cf}(\Gamma) > \aleph_0$	$E_\Gamma(V) = C_\Gamma(V) \subsetneq V_\Gamma$

Proof. A nonzero ordered group has no largest element, so its cofinality is infinite: any finite subset has a maximum, and adding a positive element gives a strictly larger group element. If the cofinality is countable, choose a cofinal enumeration $c_1, c_2, \dots$ and fix any $\eta > 0$. Define

$$\gamma_1 = c_1 + \eta, \quad \gamma_{n+1} = \max(\gamma_n, c_{n+1}) + \eta.$$

This sequence is strictly increasing and cofinal, since $\gamma_n > c_n$; no assumption that η is an order unit is used. Choose independent coefficients e_n and form $x = \sum_{n \geq 1}^H t^{\gamma_n} e_n$. For any cut γ , cofinality supplies N with $\gamma_N \geq \gamma$, and only indices $n < N$ lie below that cut. Thus its coefficient rank below each cut is finite, while its total coefficient rank is infinite. Thus it belongs to $C_\Gamma(V) \setminus E_\Gamma(V)$.

Conversely, suppose the group has uncountable cofinality and $x \in C_\Gamma(V)$. If its total coefficient rank were infinite, choose countably many linearly independent coefficients of x , recursively choosing each outside the span of its predecessors. Their distinct exponents form a countable subset of Γ . Uncountable cofinality says this subset is not cofinal, so there is $\gamma \in \Gamma$ strictly above every one of its elements. The finite-dimensional space $V_{x, < \gamma}$ would then contain all of the chosen independent coefficients, a contradiction. So $x \in E_\Gamma(V)$. Combine this with Theorem 4.5 to obtain the displayed diagram. $\square$

Thus an algebraic scalar extension can already be valuation-complete while remaining strictly smaller than the full Hahn space. Completion and closure under strong sums are genuinely different operations. For $\Gamma = 0$, the entire construction simply reduces to V over k .

Remark 4.7 (Order units versus countable cofinality). The sufficient condition in the last clause of Theorem 4.5 is the existence of an order unit; the exact condition in Theorem 4.6 is countable cofinality. These differ: the entire-functions report exhibits a group of countable cofinality with no order unit (`ent:rem:one-name`). For such a group the proof of Theorem 4.6 still produces a vector in $C_\Gamma(V) \setminus E_\Gamma(V)$ from any increasing cofinal sequence, although no single monomial t^α has cofinal powers. The same invariant decides, in that report, whether a strongly entire power series can fail to be a polynomial (`ent:cor:cofinality`); the two theorems concern different objects and neither is used to prove the other.

5 Completeness and extension: the classical mechanism

5.1 Spherical completeness without a norm on the coefficients

The extension argument needs full Hahn series, not merely a field of left-finite or bounded-denominator series. The following proof makes this distinction transparent.

Proposition 5.1 (Spherical completeness of the full Hahn space). *Every nested set-indexed family of nonempty closed valuation balls in V_Γ has nonempty intersection. In particular, K is spherically complete and V_Γ is complete for its valuation uniformity. No completeness or topology on V is required.*

Proof. Write the nested family as $B(a_j, \rho_j)$. Membership in that ball means equality with a_j at every exponent $\delta < \rho_j$. For any exponent below at least one radius, define x_δ to be that prescribed coefficient. Nested balls have compatible prescriptions: take a point in the smaller ball, which also belongs to the larger one. The ultrametric inequality then gives $v(a_j - a_k) \geq \min(\rho_j, \rho_k)$, so the centers agree at every exponent prescribed by both balls. At exponents below no radius, put $x_\delta = 0$.

We must verify that x is a Hahn vector, not just a function. Let A be a nonempty subset of its support and select $\delta_0 \in A$. A nonzero coefficient at δ_0 was prescribed by some radius $\rho_j > \delta_0$. All coefficients below ρ_j agree with those of a_j . The nonempty set $A \cap (-\infty, \rho_j)$ is therefore a subset of a well-ordered support and has a least element. That element is also the least of A , since every element not in the intersection is larger. Thus $\text{supp } x$ is well ordered. By construction x is in every ball.

For completeness, let (x_i) be a Cauchy net. For each cut γ there is i_γ such that $v(x_i - x_j) \geq \gamma$ whenever $i, j \geq i_\gamma$. Thus the entire truncation below γ , not just each individual coefficient, is constant on that tail. Truncations at two cuts are compatible by choosing a common later net index. Every exponent lies below some cut when $\Gamma \neq 0$, so these data prescribe all coefficients of x . A nonempty subset of $\text{supp } x$ has a least element by the same argument as above: choose a cut above one of its elements and use a stabilized Hahn truncation there. Hence x is a Hahn vector. For every γ , eventually $v(x_i - x) \geq \gamma$; raising γ proves convergence in each open neighborhood. For $\Gamma = 0$ the uniformity is discrete and every Cauchy net is eventually constant. The zero coefficient space is immediate; an empty family of balls also has the whole space as its intersection. $\square$

Remark 5.2. In particular, spherical completeness here does not recover the ordinary Hilbert topology on constants. Even an incomplete ordinary normed coefficient space gives a complete Hahn space for this valuation. The completeness is completeness of formal truncations.

Within the collection, the spectral report proves metric completeness of $V((t^\Gamma))$ for a real-valued metric when $\Gamma \subseteq \mathbb{R}$ (`ihh:hh:prop:complete`), and the rank-one Berkovich report proves spherical completeness of the scalar field $\mathbb{C}((t^\mathbb{R}))$ (its `prop:spherical`, a label without report prefix). Proposition 5.1 contains both for coefficient vector spaces and arbitrary rank; its proof is the same stabilization-of-truncations argument.

5.2 A valuation Hahn–Banach lemma

We record the exact extension statement used, with a proof that works for arbitrary ordered groups. The rank-one version is classical Ingleton theory [2, 3]; the argument is the same nested-ball argument, written additively.

Theorem 5.3 (Contractive scalar extension). *Let X be a K -vector space with $v(0) = +\infty$ and $v(x) \in \Gamma$ for $x \neq 0$, satisfying the scalar and ultrametric laws. Let $M \leq X$ and $f : M \rightarrow K$ be K -linear with $v(fm) \geq v(m)$ for every $m \in M$. There is an extension $L : X \rightarrow K$ with the same inequality.*

Proof. First extend over one vector $y \notin M$. For each $m \in M$ consider

$$B_m = B(f(m), v(y - m)) \subseteq K.$$

These balls form a chain. Indeed,

$$v(f(m) - f(n)) \geq v(m - n) \geq \min\{v(y - m), v(y - n)\}.$$

Write $r_m = v(y - m)$. If $r_m \leq r_n$ and $z \in B_n$, then

$$v(z - f(m)) \geq \min\{v(z - f(n)), v(f(n) - f(m))\} \geq r_m,$$

so $B_n \subseteq B_m$. Each ball contains its center, and its radius lies in Γ because $y \notin M$. Spherical completeness now provides $c \in \bigcap_{m \in M} B_m$. Define $g(m + ay) = f(m) + ac$.

The decomposition is unique because $M \cap Ky = \{0\}$. Thus g is well-defined and K -linear, and agrees with f on M . For $a \neq 0$, put $n = -m/a \in M$. Since $c \in B_n$,

$$v(g(m + ay)) = v(a) + v(c - f(n)) \geq v(a) + v(y - n) = v(m + ay).$$

The case $a = 0$ is the original bound.

Finally order all contractive extensions of f on intermediate subspaces of X by extension. Their graphs form a set of subsets of $X \times K$, and f itself is one member. Along a nonempty chain, the union of domains is a subspace and the maps agree on overlaps. Every finite linearity identity and every valuation bound occurs in one member, so the union is again a contractive extension. The empty chain has upper bound f . Zorn's lemma supplies a maximal one. If its domain were not X , the one-step argument would enlarge it. Thus its domain is all of X . $\square$

The maximal-extension argument uses Zorn's lemma and hence choice. Choosing a single element of the already proved nonempty intersection in the one-step argument does not itself require an additional choice principle. Neither step specifies a canonical or effective extension for an arbitrary input space. In particular, the continuous functionals constructed from this theorem below are exact existence results, not computable symbolic formulas on all Hahn vectors.

6 The continuous dual and its invisible kernel

6.1 The restriction space

Let

$$D_{\text{cont}}(V_\Gamma) = \text{Hom}_K^{\text{cont}}(V_\Gamma, K), \quad D_{\text{str}}(V_\Gamma) = \text{Hom}_K^{\text{str}}(V_\Gamma, K).$$

Let $V^\# = \text{Hom}_k(V, k)$ be the ordinary *algebraic* dual, not a topological dual. By Theorem 3.2,

$$D_{\text{str}}(V_\Gamma) \simeq V^\#((t^\Gamma)). \quad (6.1)$$

Definition 6.1. The restriction space $P_\Gamma(V)$ consists of the k -linear maps $A : V \rightarrow K$ such that there exists $s \in \Gamma$ with $v(Au) \geq s$ for every $u \in V$. Its K -vector-space structure is pointwise scalar multiplication.

This is indeed a K -vector space: if A, B have common bounds s, r , then $A + B$ has bound $\min(s, r)$, and for $a \neq 0$ the map aA has bound $v(a) + s$. The zero map has every bound, covering $a = 0$. All these maps remain k -linear.

An equivalent coefficient description is a family $(\varphi_\gamma)_{\gamma \in \Gamma}$ in $V^\#$ such that $\{\gamma : \varphi_\gamma(u) \neq 0\}$ is well ordered for each u , and $\bigcup_u \{\gamma : \varphi_\gamma(u) \neq 0\}$ is bounded below. The value of $A(u)$ is the associated scalar Hahn series. Conversely, these pointwise support conditions define $A(u)$, and linearity of every φ_γ proves k -linearity by coefficient comparison; the common support bound gives the required valuation bound. In contrast, $V^\#((t^\Gamma))$ requires this union itself to be well ordered. The two requirements agree for a cyclic group, but not in general.

Lemma 6.2 (Extension from constants to their algebraic span). *Every $A \in P_\Gamma(V)$ has a unique K -linear extension $f_A : E_\Gamma(V) \rightarrow K$. If $v(Au) \geq s$ for all u , then $v(f_A e) \geq v(e) + s$ for all $e \in E_\Gamma(V)$.*

Proof. Uniqueness and well-definedness follow from $E_\Gamma(V) \simeq K \otimes_k V$. To check the bound, choose linearly independent constants $u_1, \dots, u_d$ with $e = \sum_j a_j u_j$. For $e \neq 0$, omit zero scalars and put $\mu = \min_j v(a_j)$. The coefficient at μ is $\sum_j [t^\mu] a_j u_j$, a nontrivial combination of independent vectors, so cannot vanish. All coefficients below μ are zero. Thus $v(e) = \mu$, even when several scalars share their least exponent. Therefore

$$v(f_A e) = v\left(\sum_j a_j A(u_j)\right) \geq \min_j \{v(a_j) + s\} = v(e) + s.$$

The zero vector is automatic. □

Theorem 6.3 (Exact restriction sequence). *Define*

$$N_\Gamma(V) = \{L \in D_{\text{cont}}(V_\Gamma) : L(u) = 0 \text{ for every } u \in V\}.$$

Restriction to constants yields an exact sequence of K -vector spaces

$$0 \longrightarrow N_\Gamma(V) \longrightarrow D_{\text{cont}}(V_\Gamma) \xrightarrow{\text{res}} P_\Gamma(V) \longrightarrow 0. \quad (6.2)$$

Restriction is injective on $D_{\text{str}}(V_\Gamma)$; its image there is exactly the embedded copy of $V^\#((t^\Gamma))$.

Proof. A continuous functional has a uniform valuation-shift bound by Lemma 2.3. On nonzero constants the input valuation is zero, so its restriction belongs to $P_\Gamma(V)$. The kernel is exactly $N_\Gamma(V)$ by definition.

For surjectivity, let $A \in P_\Gamma(V)$ and take a lower bound s . By Lemma 6.2, $t^{-s}f_A$ is contractive on $E_\Gamma(V)$. Extend it to V_Γ by Theorem 5.3 and multiply the extension by t^s . The result is continuous and restricts to A . Finally, the assertion about strong maps is exactly Theorem 3.2, with $W = k$. $\square$

As vector spaces, the exact sequence admits a noncanonical splitting by choice. There is no specified canonical splitting and no assertion of a preferred extension operation. The sequence describes all restrictions and all ambiguities, rather than concealing the ambiguity in a purported coefficient formula for every continuous functional.

6.2 Invisible functionals detect exactly the completion defect

We call members of $N_\Gamma(V)$ *invisible on constants*. This is a definition, not a statement that they are zero: the constants need not be topologically total in the Hahn space.

Theorem 6.4 (Separation of the completion defect). *For every V and every nonzero Γ ,*

$$C_\Gamma(V) = \bigcap_{L \in N_\Gamma(V)} \ker L. \quad (6.3)$$

More explicitly, if $x \notin C_\Gamma(V)$ and $\dim_k V_{x, < \gamma} = \infty$, there is a K -linear functional L satisfying

$$L|_V = 0, \quad L(x) = t^\gamma, \quad v(Ly) \geq v(y) \quad (y \in V_\Gamma). \quad (6.4)$$

Such a functional is continuous but not strong.

Proof. Every member of $N_\Gamma(V)$ vanishes on the algebraic span of the constants and, by continuity, on its closure. This proves one inclusion in (6.3).

For the other, if $x \notin C_\Gamma(V)$, the completion criterion Theorem 4.2 supplies a γ with $\dim_k V_{x, < \gamma} = \infty$. No $e \in E_\Gamma(V)$ can agree with x below γ : all coefficients of e lie in one finite-dimensional space, which would then contain $V_{x, < \gamma}$. Thus

$$v(x - e) < \gamma \quad (e \in E_\Gamma(V)). \quad (6.5)$$

Since $x \notin E_\Gamma(V)$, the decomposition $e + ax$ is unique: a nonzero difference between two scalar coefficients would express x as a member of $E_\Gamma(V)$. Thus

$$f(e + ax) = at^\gamma$$

is a well-defined K -linear map on $E_\Gamma(V) + Kx$. If $a \neq 0$, apply (6.5) to $-a^{-1}e \in E_\Gamma(V)$ to obtain

$$v(e + ax) = v(a) + v(x + a^{-1}e) < v(a) + \gamma = v(f(e + ax)).$$

If $a = 0$, the output is zero. Hence f is contractive in valuation. Apply Theorem 5.3 to extend it to L on all of V_Γ . This proves (6.4). In particular, $L(x) = t^\gamma \neq 0$, so x lies outside the common kernel.

Finally $x = \sum_\delta^H t^\delta x_\delta$, while $L(t^\delta x_\delta) = 0$ for every monomial term. Those images are a summable zero family, but their sum is not $L(x)$. Therefore L is not strong. $\square$

Corollary 6.5 (Exact existence boundary). *For nonzero Γ , invisible nonzero continuous functionals exist if and only if V is infinite-dimensional and Γ is not ordered isomorphic to $\mathbb{Z}$. Moreover $N_\Gamma(V) \cap D_{\text{str}}(V_\Gamma) = \{0\}$.*

Proof. By Theorem 6.4, the kernel is nonzero exactly when the algebraic scalar extension is not dense. Apply Theorem 4.5. A strong functional vanishing on constants vanishes on every monomial coefficient expansion, proving the intersection statement. $\square$

7 Two independent ways continuity can fail strongness

7.1 One failure changes sums; the other destroys summability

The invisible functionals just constructed annihilate every term of a particular strong sum but not its value. A different construction produces an image family that is not even summable.

Theorem 7.1 (Two failures, for every noncyclic group). *Assume $\dim_k V = \infty$ and $\Gamma \neq 0$ is not ordered isomorphic to $\mathbb{Z}$. There exist continuous K -linear functionals of both following types.*

(a) *A nonzero functional is zero on all constants and fails commutation with a strong monomial sum whose termwise images are all zero.*

(b) *A functional has a pointwise finite-support restriction to constants, bounded below in valuation, but sends some strongly summable monomial family to the constant family $(1)_{n \geq 1}$. It cannot be written as a strong functional plus an invisible one.*

Proof. Part (a) is Theorem 6.4 and Corollary 6.5. For (b), Lemma 4.4 provides an increasing sequence $\gamma_n < \beta$. Put $\delta_n = \beta - \gamma_n$; these are strictly decreasing and positive. Choose independent vectors e_n and extend them to a Hamel basis of V . Define $A : V \rightarrow K$ by

$$A(e_n) = t^{\delta_n}, \quad A(b) = 0 \quad \text{on the other basis vectors.}$$

Every vector is a finite Hamel combination, so $A(u)$ has finite support. All those supports are positive, so $A \in P_\Gamma(V)$ with lower bound zero. By Theorem 6.3, extend A to a continuous K -linear functional L on V_Γ .

The family $t^{-\delta_n} e_n$ is strongly summable, its exponents being strictly increasing. But

$$L(t^{-\delta_n} e_n) = t^{-\delta_n} A(e_n) = 1 \quad (n \geq 1).$$

The image family violates the finite-fiber condition at zero: infinitely many nonzero coefficients occur there. This obstruction holds in every characteristic; it does not involve assigning a value to an infinite sum of ones. If $L = S + N$ with S strong and N invisible on constants, then $S|_V = A$. This is impossible by Theorem 3.2, because the global coefficient support of A contains the strictly decreasing sequence δ_n . $\square$

Corollary 7.2 (A second quotient obstruction). *Restriction induces a canonical isomorphism*

$$\frac{D_{\text{cont}}(V_\Gamma)}{D_{\text{str}}(V_\Gamma) + N_\Gamma(V)} \simeq \frac{P_\Gamma(V)}{V^\#((t^\Gamma))}. \quad (7.1)$$

Under the hypotheses of Theorem 7.1, this quotient is nonzero, as is $N_\Gamma(V)$ itself.

Proof. Write $Q = V^\#((t^\Gamma))$ for the identified subspace of $P_\Gamma(V)$. The K -linear map

$$L \mapsto \text{res}(L) + Q$$

is onto by Theorem 6.3. Its kernel contains $D_{\text{str}} + N_\Gamma(V)$. Conversely, if $\text{res}(L) \in Q$, the strong-map theorem gives a strong S with the same restriction, and $L - S \in N_\Gamma(V)$. The kernel is therefore exactly that sum, and the first isomorphism theorem gives the displayed canonical *algebraic* quotient isomorphism; no quotient topology or splitting is chosen. Part (b) gives a nonzero quotient class, while part (a) gives a nonzero member of $N_\Gamma(V)$. $\square$

In part (a), preservation of summability is asserted only for the displayed zero image family, not for every possible strong family. The two examples give distinct obstructions; they do not assert logical independence of the two universally quantified clauses in the definition.

Thus the full continuous dual is not obtained simply by adding invisible functionals to the strong dual. There are also continuous restrictions whose supports cannot be organized as one Hahn series of coefficient functionals.

7.2 Why ordinary Laurent exponents are exceptional

Theorem 7.3 (When continuous and strong duals agree). *For $\Gamma \neq 0$,*

$$D_{\text{cont}}(V_\Gamma) = D_{\text{str}}(V_\Gamma) \iff (\dim_k V < \infty \text{ or } \Gamma \simeq_{\text{ord}} \mathbb{Z}).$$

Proof. If V is finite-dimensional, a k -basis $u_1, \dots, u_d$ of V , viewed as constant Hahn vectors, is a K -basis of V_Γ . For any K -linear functional L , the finite union $\bigcup_{j=1}^d \text{supp } L(u_j)$ is well ordered. Define $\varphi_\gamma(u_j) = [t^\gamma]L(u_j)$ and extend k -linearly. These coefficient functionals form one Hahn series, whose strong map agrees with L on the K -basis and hence everywhere. The empty basis also covers $V = 0$.

If $\Gamma \simeq_{\text{ord}} \mathbb{Z}$, then $C_\Gamma(V) = V_\Gamma$, so $N_\Gamma(V) = 0$. For any $A \in P_\Gamma(V)$, its global support is bounded below in an ordered copy of $\mathbb{Z}$, and is therefore well ordered. Hence $P_\Gamma(V) = V^\#((t^\Gamma))$. The exact sequence (6.2) now identifies every continuous functional with the unique strong functional having its restriction.

In every remaining case, Theorem 7.1 supplies continuous nonstrong functionals. $\square$

There is a useful topological explanation in the cyclic case. For a strong family $(x_j)_{j \in J}$ and a cut γ , put

$$J_\gamma = \{j \in J : \text{supp } x_j \cap (-\infty, \gamma) \neq \emptyset\}.$$

Its well-ordered support union is bounded below in an ordered copy of $\mathbb{Z}$, so it has only finitely many exponents below γ . Each exponent receives finitely many contributions; hence J_γ is finite. Finite subsums containing J_γ agree with the strong sum below γ . Taking γ above any prescribed valuation cutoff proves convergence of the net indexed by finite subsets of J .

If L has shift bound s , an image with support below γ can only come from an index in $J_{\gamma-s}$. Also the whole image support is bounded below by a lower bound of the input support plus s (the all-zero family is immediate). In the cyclic group that support is well ordered, and the preceding finite-index property gives finite fibers. The image family is therefore strong, and its finite subsums converge to its strong sum. By linearity and continuity they also converge to $L(\sum_j^H x_j)$; uniqueness of limits proves equality. For a noncyclic group, the bounded increasing sequence used above prevents this finite-cut argument.

For the excluded zero group, the valuation topology is discrete and strong families have only finitely many nonzero members. Thus $D_{\text{cont}} = D_{\text{str}} = P_0(V) = V^\#$ and $N_0(V) = 0$.

8 Hilbert coefficients: represented, strong, and bounded duals

8.1 The positive form and its field-valued norm

Take $k = \mathbb{C}$, an ordinary complex Hilbert space H , and

$$K = \mathbb{C}((t^\Gamma)), \quad F = \mathbb{R}((t^\Gamma)), \quad H_\Gamma = H((t^\Gamma)).$$

Order F by its first nonzero coefficient and conjugate K coefficientwise. Our inner products are linear in the first variable. The coefficient-convolution form, already used in the repository [9] as `ihh:hh:eq:inner` with the same linearity convention, is

$$\langle x, y \rangle = \sum_{\alpha, \beta}^H \langle x_\alpha, y_\beta \rangle_H t^{\alpha+\beta}. \quad (8.1)$$

Each ordinary Hilbert inner product is evaluated as one coefficient *before* applying Hahn convolution. There is no strong summation over infinitely many ordinary coordinates at the same exponent.

Proposition 8.1 (Norm and valuation; standard coefficient extension). *The form (8.1) is positive definite. Its positive square-root norm $\|x\| = \sqrt{\langle x, x \rangle}$ exists in F for every ordered group Γ , satisfies Cauchy–Schwarz and the triangle inequality, and*

$$v(\|x\|) = v(x) \quad (x \neq 0).$$

If $\delta = v(x)$, its leading coefficient is $\|x_\delta\|_H$.

This is the spectral report’s `ihh:hh:prop:inner`, which since its revision 6e34651 also holds without divisibility of Γ . The proof is repeated so that this report can be read alone.

Proof. Convolution is defined by Lemma 2.2. If $x \neq 0$, the first coefficient of $\langle x, x \rangle$ is $\|x_\delta\|_H^2 > 0$ at exponent 2δ . Write it as $c^2 t^{2\delta}(1+u)$ with $c > 0$ and $v(u) > 0$, allowing $u = 0$. Then

$$\|x\| = ct^\delta \sum_{n \geq 0}^H \binom{1/2}{n} u^n$$

is defined by the Hahn–Neumann lemma and squares to the form value. Its first coefficient and exponent are as asserted. Notice that δ is already in Γ : divisibility of Γ is not needed for these particular square roots. The case $H = \mathbb{C}$ defines $|a| = \sqrt{a\bar{a}}$ for scalars as well. Set $\|0\| = 0$. Sesquilinearity gives $\langle ax, ax \rangle = |a|^2 \langle x, x \rangle$; uniqueness of the nonnegative square root then gives $\|ax\| = |a|\|x\|$, including the zero cases.

Completing the square in $\langle x - ay, x - ay \rangle \geq 0$, with $a = \langle x, y \rangle / \langle y, y \rangle$ for $y \neq 0$, proves $|\langle x, y \rangle|^2 \leq \langle x, x \rangle \langle y, y \rangle$. Taking nonnegative square roots gives Cauchy–Schwarz, with $y = 0$ immediate. For $z \in K$, write $z = \Re z + i\Im z$ with real Hahn components. The identity $|z|^2 = (\Re z)^2 + (\Im z)^2$ gives $\Re z \leq |z|$. Consequently

$$\|x + y\|^2 = \|x\|^2 + 2\Re \langle x, y \rangle + \|y\|^2 \leq (\|x\| + \|y\|)^2,$$

which gives the triangle inequality by monotonicity of nonnegative squaring in the ordered field F . These manipulations require only the square roots already constructed. $\square$

For $\Gamma \neq 0$, the topology defined by all positive F -valued norm tolerances agrees with the intrinsic valuation topology. Indeed, write $B_\epsilon = \{x : \|x\| < \epsilon\}$ for $\epsilon > 0$. The leading-exponent comparison gives

$$U_{v(\epsilon)}(0) \subseteq B_\epsilon, \quad B_{t^{\gamma+\eta}} \subseteq U_\gamma(0) \quad (\gamma \in \Gamma, \eta > 0).$$

For the second inclusion, $\|x\| < t^{\gamma+\eta}$ forces $v(x) \geq \gamma + \eta > \gamma$ for $x \neq 0$; the first is the same comparison in the other direction. Translation gives equality of the topologies. This statement concerns the *fixed* field F , not the topology induced from the full surreal class.

8.2 Boundedness is much weaker than ordinary coefficient boundedness

Proposition 8.2 (Order bounds for all continuous maps). *For $\Gamma \neq 0$ and ordinary Hilbert spaces H, G , a K -linear map $T : H_\Gamma \rightarrow G_\Gamma$ is valuation-continuous if and only if there is $M \in F_{>0}$ such that*

$$\|Tx\| \leq M\|x\| \quad (x \in H_\Gamma). \quad (8.2)$$

In particular, every strong map is bounded in this sense, even if its coefficient maps are ordinary discontinuous algebraic maps. If T is strong with least operator exponent s , then for each $\eta > 0$ the scalar $M = t^{s-\eta}$ is a valid bound.

Proof. Continuity gives $v(Tx) \geq v(x) + s$ for some s by Lemma 2.3. The valuation of $t^{s-\eta}\|x\|$ is strictly smaller than that of $\|Tx\|$ when both are nonzero. In an ordered Hahn field this means the former positive element is larger, irrespective of their ordinary leading coefficients. The zero cases are immediate. Conversely, a positive inequality (8.2) implies $v(Tx) \geq v(M) + v(x)$: otherwise its left side would dominate its right side. Apply Lemma 2.3. For strong maps use Corollary 3.3. $\square$

For a nonzero adjointable operator the spectral report gives the sharper bound $M = ct^{\gamma_0}$ with an ordinary real $c > \|A_{\gamma_0}\|_H$ (`ihshh:cor:autobounded`). For a strong map whose leading coefficient A_s is ordinarily unbounded, the positive bounds are instead exactly those with $v(M) < s$, and need not be monomials. Sufficiency is the strict valuation comparison in the preceding proof. For necessity, test on constant u . If $v(M) > s$, any u with $A_s u \neq 0$ violates the bound. If $v(M) = s$ and its positive leading coefficient is c , unboundedness supplies u with $\|A_s u\|_G > c\|u\|_H$, again violating it at the leading coefficient.

We do not define a real-Hahn operator norm by taking the least bound: the set of bounds need not have a least element. For the coefficientwise extension of an ordinary bounded operator on *nonzero* H , the spectral report considers *nonnegative* bounds, allowing $M = 0$. Their set has a least element precisely when the ordinary operator norm is attained (`ihshh:thm:norm`). The distinction from our positive-bound convention matters for the zero operator: its least nonnegative bound is zero, while its positive bounds have no least member, since each can be halved. For a nonzero operator zero is not a bound, so the conventions agree. The following theorem treats an ordinary *unbounded* coefficient functional and describes its bounds exactly.

Theorem 8.3 (An ordinary unbounded functional has precisely infinite bounds). *Let $\varphi : H \rightarrow \mathbb{C}$ be an ordinary unbounded algebraic linear functional, and let $\widehat{\varphi}$ be its coefficientwise Hahn extension. For $M \in F_{>0}$,*

$$|\widehat{\varphi}(x)| \leq M\|x\| \quad \text{for all } x \iff v(M) < 0.$$

Thus the admissible bounds are exactly the positive infinite Hahn scalars, and this set has no least element.

Proof. Coefficientwise extension has $v(\widehat{\varphi}x) \geq v(x)$. Every positive M with $v(M) < 0$ therefore dominates the output/input norm ratio, proving sufficiency. If $v(M) \geq 0$, the scalar is finite, so $M < c$ for some positive ordinary real c . Testing the proposed bound on constant vectors would give $|\varphi(u)| \leq c\|u\|_H$ for all u , contradicting unboundedness. Since $\Gamma \neq 0$, choosing $\eta > 0$ gives at least one such bound $t^{-\eta}$. Finally $M/2$ is a smaller positive infinite bound whenever M is one, so this nonempty set has no least element. $\square$

Such φ exist for every infinite-dimensional Hilbert space. Choose an orthonormal sequence e_n , prescribe $\varphi(e_n) = n$ on its algebraic span, and extend the sequence to a Hamel basis, prescribing zero on the added basis vectors. The basis extension uses choice. The unit vectors e_n prove unboundedness. No continuity of this ordinary functional is asserted.

8.3 Exactly which strong functionals have representers?

Write $H^\# = \text{Hom}_{\mathbb{C}}(H, \mathbb{C})$ and let $H' \subset H^\#$ be the ordinary continuous dual. Define

$$D_{\text{rep}} = \{L_y : x \mapsto \langle x, y \rangle ; y \in H_\Gamma\}.$$

This is a K -subspace of the scalar-valued functionals: with our convention $aL_y = L_{\bar{a}y}$. The map $y \mapsto L_y$ is conjugate linear, not linear. This convention does not affect the following linear description in terms of coefficient functionals.

Theorem 8.4 (Coefficientwise Riesz criterion and exact first gap). *There are canonical identifications of spaces of functionals*

$$D_{\text{str}} \simeq H^\#((t^\Gamma)), \quad (8.3)$$

$$D_{\text{rep}} \simeq H'((t^\Gamma)), \quad (8.4)$$

$$D_{\text{str}}/D_{\text{rep}} \simeq (H^\#/H')((t^\Gamma)). \quad (8.5)$$

A strong functional is inner-product-represented exactly when every coefficient functional in its global Hahn expansion is ordinarily continuous. Its representer is unique.

Proof. The first identification is Theorem 3.2. For a given $y \in H_\Gamma$, put $\varphi_\gamma(u) = \langle u, y_\gamma \rangle_H$. Each functional is ordinarily continuous, and is nonzero exactly when $y_\gamma \neq 0$ (test $u = y_\gamma$). Thus their common support is $\text{supp } y$, a well-ordered set. The strong-map theorem constructs the corresponding strong functional, and its convolution formula is exactly $L_y(x) = \langle x, y \rangle$. This proves both that represented functionals are strong and that their coefficients lie in H' . Conversely, let $L = \sum_\gamma^\text{H} \varphi_\gamma t^\gamma$ be strong with each $\varphi_\gamma \in H'$. The ordinary Hilbert-space Riesz representation theorem gives a unique $y_\gamma \in H$ satisfying $\varphi_\gamma(u) = \langle u, y_\gamma \rangle_H$. A coefficient functional is nonzero exactly when its representer is nonzero, so these y_γ have the same well-ordered support. Their Hahn vector y satisfies $L(x) = \langle x, y \rangle$ by convolution. If $L_y = L_z$, testing on $x = y - z$ gives $\langle y - z, y - z \rangle = 0$, so positive definiteness proves uniqueness.

For the quotient, apply coefficientwise the quotient map $q : H^\# \rightarrow H^\#/H'$. It defines a K -linear map on Hahn spaces whose kernel is exactly $H'((t^\Gamma))$. It is onto: choose a $\mathbb{C}$ -linear section of q and lift every coefficient through that section, without enlarging the support. The quotient map and resulting isomorphism are canonical; the auxiliary section need not be. $\square$

This coefficientwise Riesz criterion is consistent with the repository's classification of adjointable operators [9]. For L_y , the everywhere-defined K -linear map $B(a) = ay$ is an adjoint, since

$$\langle L_y x, a \rangle_K = \langle x, y \rangle \bar{a} = \langle x, ay \rangle, \quad \langle a, b \rangle_K = a \bar{b}.$$

Conversely, an adjoint B satisfies $Lx = \langle Lx, 1 \rangle_K = \langle x, B1 \rangle$, so $L = L_{B1}$. These direct checks supply the functional case; the spectral report states `ihh:hh:thm:adjoint` for endomorphisms of H_Γ . We do not claim to have discovered a general failure of Riesz representation in non-Archimedean analysis. What the formula adds is an explicit description of the entire first gap for this Hahn coefficient model.

Remark 8.5 (Editorial: the spectral report's Riesz counterexample lies in the first gap). The spectral report constructs a nonzero valuation-continuous functional on H_Γ , for $H = \ell^2$, that vanishes on the range of its coercive operator $C = D^2 + s^2 I$ and is not inner-product-represented (`ihh:hh:prop:riesz`; it is called F there, and we write Φ since F is our real Hahn field). It has the form $\Phi = \psi_\Gamma \Pi$, where $\Pi = I - C(I + UE)^{-1}U$ is built from an algebraic left inverse U of D^2 , $E = s^2 I$, and ψ is a nonzero linear functional on an algebraic complement W of $\text{Ran } D^2$. All coefficients of Π are ordinary algebraic endomorphisms of H , and $\psi_\Gamma \Pi = (\psi \pi_W)_\Gamma \Pi$ with π_W the algebraic projection onto W along $\text{Ran } D^2$. By Theorem 3.2 both factors are strong, so $\Phi \in D_{\text{str}}$. Its coefficient at exponent zero is $\psi \pi_W$, because the zero coefficient of Π is $I - D^2 U = \pi_W$. This functional vanishes on the dense subspace $\text{Ran } D^2$ of ℓ^2 and is nonzero, so it is not ordinarily continuous. Hence the class of Φ in (8.5) is nonzero: `ihh:hh:prop:riesz` is an explicit element of the first gap $D_{\text{str}} \setminus D_{\text{rep}}$, not of the second gap $D_{\text{cont}} \setminus D_{\text{str}}$. This is consistent with that proposition holding for every nonzero Γ , including $\Gamma \simeq_{\text{ord}} \mathbb{Z}$, where Corollary 8.6 below gives $D_{\text{str}} = D_{\text{cont}}$. The remark is an editorial comparison added when this report was placed; it uses only the two cited statements.

Corollary 8.6 (The three-dual trichotomy). *Assume $\Gamma \neq 0$. If H is finite-dimensional, then $D_{\text{rep}} = D_{\text{str}} = D_{\text{cont}}$. If H is infinite-dimensional and $\Gamma \simeq_{\text{ord}} \mathbb{Z}$, then*

$$D_{\text{rep}} \subsetneq D_{\text{str}} = D_{\text{cont}}.$$

If H is infinite-dimensional and Γ is noncyclic, then

$$D_{\text{rep}} \subsetneq D_{\text{str}} \subsetneq D_{\text{cont}}. \tag{8.6}$$

Every functional in these three spaces is bounded by some positive F -scalar.

Proof. In finite dimension every ordinary functional is continuous and every K -linear functional has a finite coordinate representation. In infinite dimension an unbounded algebraic functional supplies the first strict inclusion by Theorem 8.4. The second equality or strict inclusion follows from Theorem 7.3. Boundedness is Proposition 8.2. $\square$

8.4 A cardinal lower bound on the first gap

This optional quantitative statement shows that the first gap is not accounted for by a few exceptional discontinuous coefficients.

Proposition 8.7. *If H is infinite-dimensional and separable and $\mathfrak{c} = 2^{\aleph_0}$, then*

$$\dim_K(D_{\text{str}}/D_{\text{rep}}) \geq 2^{\mathfrak{c}}.$$

No equality is asserted for arbitrary Γ .

Proof. An infinite-dimensional separable complex Hilbert space has cardinality $\mathfrak{c}$ and Hamel dimension $\mathfrak{c}$. For the lower bound on that dimension, fix an orthonormal sequence and use the vectors $u_a = \sum_{n \geq 0} a^n e_n$, $0 < a < 1$. Every finite selection is linearly independent, by the Vandermonde determinant obtained from its first coordinates. The cardinal upper bound follows from separability, since vectors are determined by countably many complex coordinates in an orthonormal basis.

The algebraic dual consists of all assignments of complex values to a Hamel basis, so has cardinality $\mathfrak{c}^\mathfrak{c} = 2^\mathfrak{c}$. A vector space over a field of size $\mathfrak{c}$ has cardinality equal to the maximum of its infinite dimension and $\mathfrak{c}$. Consequently $\dim_{\mathbb{C}} H^\# = 2^\mathfrak{c} > \mathfrak{c}$. Riesz is conjugate-linear with our convention, but preserves independence and spanning, so $\dim_{\mathbb{C}} H' = \dim_{\mathbb{C}} H = \mathfrak{c}$. Extend a basis of H' to one of $H^\#$. The added basis vectors must number $2^\mathfrak{c}$: the union of two sets of smaller cardinality, one of size $\mathfrak{c}$, cannot have cardinality $2^\mathfrak{c}$. Their quotient classes form a basis, proving $\dim_{\mathbb{C}}(H^\#/H') = 2^\mathfrak{c}$ without any continuum hypothesis.

Independent constant coefficient vectors remain independent over K . Indeed, in a nontrivial finite relation $\sum_j a_j w_j = 0$, omit zero scalars and let $\mu = \min_j v(a_j)$. At exponent μ the relation would give $\sum_j [t^\mu] a_j w_j = 0$ with at least one nonzero complex coefficient, contradicting independence. Apply this to a basis of $H^\#/H'$ and use (8.5). $\square$

9 Closed hyperplanes with no orthogonal geometry

9.1 A strong bounded projection without an orthogonal decomposition

Let $\varphi : H \rightarrow \mathbb{C}$ be an unbounded algebraic functional as above, let $M = \ker \varphi$, and choose $u \in H$ with $\varphi(u) = 1$. The ordinary algebraic hyperplane M is dense in H . Indeed, if its closure were proper, its codimension-one algebraic maximality would force $\overline{M} = M$. The ordinary Hilbert projection theorem would then give $H = M \oplus M^\perp$, with $M^\perp$ one-dimensional. For a unit vector w spanning that complement, $x - \langle x, w \rangle_H w \in M$, so

$$\varphi(x) = \varphi(w) \langle x, w \rangle_H, \quad |\varphi(x)| \leq |\varphi(w)| \|x\|_H.$$

This would contradict the choice of the unbounded φ .

Put $M_\Gamma = M((t^\Gamma)) \subset H_\Gamma$. The ordinary density of M and the valuation closedness of M_Γ are compatible because the two topologies treat coefficients completely differently.

Theorem 9.1 (A closed, strongly complemented hyperplane with zero orthogonal complement). *For every nonzero Γ and every infinite-dimensional ordinary complex Hilbert space H , the subspace M_Γ above has all the following properties. It is a valuation-closed K -hyperplane; it is closed under strong sums; and*

$$P_M(x) = x - u\hat{\varphi}(x)$$

is a strong, order-bounded projection onto it. Nevertheless

$$M_\Gamma^\perp = \{0\},$$

so there is no orthogonal projection onto M_Γ and $H_\Gamma \neq M_\Gamma \oplus M_\Gamma^\perp$.

Proof. The coefficientwise map $\hat{\varphi}$ is onto K , because $\hat{\varphi}(au) = a$. Its kernel is exactly M_Γ . It is continuous, so that kernel is closed, since K is Hausdorff; surjectivity makes its codimension one. Directly, if $x \notin M_\Gamma$, choose a coefficient $x_\delta \notin M$. Every vector in $U_\delta(x)$ agrees with x through that

coefficient and is therefore outside M_Γ . Strong sums stay in M_Γ since every coefficient sum is finite and lies in M .

The ordinary algebraic map $p = I - u\varphi$ has $p^2 = p$ and range M , because $\varphi(u) = 1$. Its coefficientwise Hahn extension is precisely P_M , so it is strong by Theorem 3.2. In fact $v(P_M x) \geq v(x)$; hence every $t^{-\eta}$ with $\eta > 0$ is an order bound, by the proof of Proposition 8.2. The decomposition is explicitly

$$x = P_M x + u\hat{\varphi}(x), \quad H_\Gamma = M_\Gamma \oplus Ku.$$

The intersection is zero since $\hat{\varphi}(au) = a$. This is an algebraic direct sum; no orthogonality has been asserted.

If $y \perp M_\Gamma$, testing against every constant $m \in M$ gives $\langle m, y_\delta \rangle_H = 0$ for every δ . Ordinary density of M forces $y_\delta = 0$ for all δ , hence $y = 0$. If an orthogonal projection Q onto M_Γ existed, then $x - Qx \in M_\Gamma^\perp = \{0\}$ for every x . It would follow that $Q = I$, contradicting that its range is the proper hyperplane M_Γ . The same argument rules out an orthogonal direct-sum decomposition. $\square$

This is stronger than saying that a particular constructive formula for an orthogonal projection fails. The subspace has codimension one and a strong bounded algebraic projection, but no orthogonal direct-sum complement; its orthogonal complement itself exists and equals zero. The general non-Archimedean phenomenon has predecessors [4]; the role of the coefficient topology is fully explicit here. (We write P_M for this projection to keep the letter P free for the restriction space $P_\Gamma(V)$.)

Inside the collection the nearest predecessor is the spectral report's closed-range theorem (`ihh:hh:thm:closedrange`): in the same space H_Γ , with $H = \ell^2$, the range $M = CH_\Gamma$ of the coercive operator $C = D^2 + s^2 I$ is valuation-closed and proper with $M^\perp = \{0\}$. At the pin of the source manuscript that theorem was proved, by a metric argument, for divisible $\Gamma \subseteq \mathbb{R}$; the spectral report now proves closedness at every rank, as the kernel of a valuation-continuous projection whose coefficients are ordinary algebraic maps. The two witnesses differ in codimension: that range has algebraic cokernel of K -dimension at least $2^{\aleph_0}$ (`ihh:hh:thm:continuum`), whereas M_Γ here is a hyperplane, the kernel of one strong functional. Both are kernels of strong idempotents in the sense of Theorem 3.2.

9.2 The distance does not merely fail to attain a minimum

Theorem 9.2 (Exact lower-bound cut for the distance). *Assume $\Gamma \neq 0$ and retain H and $M = \ker \varphi$ from the preceding theorem. For any constant $u \in H \setminus M$, consider*

$$\mathcal{D}_u = \{\|u - m\| : m \in M_\Gamma\} \subset F_{>0}.$$

Its nonnegative lower bounds are exactly zero and all positive infinitesimals of F . Consequently $\mathcal{D}_u$ has no infimum in F .

Proof. For $m \in M_\Gamma$, its coefficient m_0 lies in M , so $u - m_0 \neq 0$. With $v(0) = \infty$, this gives the exact identity

$$v(u - m) = \min\{0, v(m)\}.$$

If $v(m) < 0$, its leading term survives with a minus sign; otherwise the nonzero coefficient $u - m_0$ at zero is leading. Thus $\|u - m\|$ is either positive infinite or has valuation zero with positive ordinary leading coefficient. In both cases every positive infinitesimal is strictly smaller than the distance.

Conversely, density of M in the ordinary Hilbert topology gives, for every ordinary $r > 0$, a constant $m \in M$ with $\|u - m\|_H < r$. A nonnegative lower bound in F must therefore be smaller than every positive real. It is either zero or positive infinitesimal. This identifies the lower-bound set exactly. It has no greatest element: choose $\eta > 0$ in Γ , so t^η is a positive infinitesimal lower bound larger than zero; and any positive infinitesimal lower bound can be doubled while remaining one. Any infimum would itself be nonnegative because zero is a lower bound. It would therefore be a greatest element of the identified set, which is impossible. $\square$

The ordinary distance from u to M is zero. The Hahn field does not replace that distance by a particular positive infinitesimal. Rather, all positive infinitesimals lie below every available distance, and there is no greatest one. Confusing an infinitesimal scale with a complete ordered continuum would miss this distinction.

10 Enlarging the exponent group can undo completion

10.1 The exact cofinality dichotomy

Let $\Gamma \subseteq \Delta$ be an ordered subgroup inclusion, and embed K_Γ in K_Δ and V_Γ in V_Δ by preserving coefficients and exponents and adding zero coefficients outside Γ . The field inclusion and the K_Γ -linear vector inclusion preserve strong sums: the same well-ordered support union and the same finite coefficient fibers remain admissible in Δ . Each group supplies its own intrinsic valuation topology. Recall that Γ is cofinal in Δ if for every $\delta \in \Delta$ some $\gamma \in \Gamma$ satisfies $\delta \leq \gamma$.

Theorem 10.1 (Cofinal preservation and noncofinal collapse). *For nonzero $\Gamma \subseteq \Delta$ and any V ,*

$$E_\Delta(V) \cap V_\Gamma = E_\Gamma(V), \quad (10.1)$$

and

$$C_\Delta(V) \cap V_\Gamma = \begin{cases} C_\Gamma(V), & \Gamma \text{ is cofinal in } \Delta, \\ E_\Gamma(V), & \Gamma \text{ is not cofinal in } \Delta. \end{cases} \quad (10.2)$$

These are intersections of explicitly embedded vector sets; no claim that a completed tensor-product functor commutes with extension is implicit.

Proof. By Proposition 4.1, algebraic scalar extension is characterized by finite total coefficient rank. The embedding adds only zero coefficients, so the coefficient span in V is unchanged. This proves (10.1).

If Γ is cofinal and $x \in C_\Gamma(V)$, then for every $\delta \in \Delta$ choose $\gamma_0 \in \Gamma$ with $\delta \leq \gamma_0$, then fix $\eta \in \Gamma_{>0}$ and put $\gamma = \gamma_0 + \eta > \delta$. The coefficients of x below δ lie among those below γ and have finite rank. Thus $x \in C_\Delta(V)$. The converse follows by testing only cuts in Γ .

If Γ is not cofinal, some $\delta \in \Delta$ is larger than every element of Γ . For $x \in V_\Gamma$, all its coefficients occur below this one cut. Membership in $C_\Delta(V)$ therefore forces finite total coefficient rank, or $x \in E_\Gamma(V)$. Conversely, $E_\Gamma(V) \subseteq E_\Delta(V) \subseteq C_\Delta(V)$, which supplies the reverse inclusion after intersecting with V_Γ . $\square$

The same dichotomy describes the induced topology. In the cofinal case, with superscripts recording the workspace,

$$U_\gamma^\Gamma(0) = U_\gamma^\Delta(0) \cap V_\Gamma, \quad U_\gamma^\Gamma(0) \subseteq U_\delta^\Delta(0) \cap V_\Gamma \quad (\gamma > \delta).$$

The first equality and cofinality make the two neighborhood systems cofinal in one another. In the noncofinal case choose $\beta \in \Delta$ above every element of Γ . Then

$$U_\beta^\Delta(0) \cap V_\Gamma = \{0\},$$

so the induced topology is discrete. If $V \neq 0$, the original topology is not discrete: fix $u \neq 0$ and $\eta \in \Gamma_{>0}$; for every old cut γ , $t^{\gamma+\eta}u \neq 0$ lies in $U_\gamma^\Gamma(0)$. Thus the new topology is strictly finer. For $V = 0$ both are discrete. If the excluded group $\Gamma = 0$ is allowed, then $E_0(V) = C_0(V) = V_0 = V$; both intersections in the theorem equal V , and its original topology is already discrete.

Remark 10.2 (Two other change-of-workspace statements in the collection). The formula (10.2) has the same shape as the entire-functions report's sharp change-of-workspace criterion $\mathcal{E}_\Delta \cap K_\Gamma[[Z]] = \mathcal{E}_\Gamma$ or $K_\Gamma[Z]$, according as Γ is or is not cofinal in Δ (`ent:thm:extension-criterion`, proved there for divisible groups). In both, the series themselves persist under embedding. A single new exponent above all of Γ removes entireness for every old nonpolynomial entire series, and excludes every old vector outside $E_\Gamma(V)$ from $C_\Delta(V)$. The objects differ (entire power series there, completed coefficient spans here), and neither theorem is used for the other. The spectral report's persistence statement (`ih:hh:cor:extension`) concerns yet another question. For a constant normal operator its spectrum in the larger field intersects the old field in its old spectrum. The same corollary preserves the stated nonsurjectivity of an injective ordinary operator under a positive-order perturbation. Both assertions hold for every enlargement, cofinal or not, whereas (10.2) distinguishes the two.

10.2 A completely explicit two-scale example

Take $\Gamma = \mathbb{Q}$ and $\Delta = \mathbb{Q} \oplus_{\text{lex}} \mathbb{Q}$, embedding $q \mapsto (0, q)$. Both groups are divisible. Let e_n be linearly independent and form

$$x = \sum_{n \geq 1}^H t^{(0,n)} e_n, \quad (10.3)$$

viewed first as $\sum_{n \geq 1}^H t^n e_n$ in $V_\mathbb{Q}$. The support is increasing and each exponent occurs once, so this family is strong in either group. There are finitely many exponents below each rational cut, so $x \in C_\mathbb{Q}(V)$. For $x_N = \sum_{n=1}^N t^n e_n$, independence in particular gives $e_{N+1} \neq 0$ and hence $v(x - x_N) = N + 1$. Given any rational cutoff, all sufficiently large N exceed it, proving convergence in the intrinsic $\mathbb{Q}$ -valuation topology. Its total coefficient rank is infinite, so $x \notin E_\mathbb{Q}(V)$.

In Δ , the single exponent $\beta = (1, 0)$ exceeds every $(0, q)$. Below this new cut, the coefficients of x have infinite rank. Hence $x \notin C_\Delta(V)$. Every finite coefficient-span approximant y , even using scalars from K_Δ , has

$$v(x - y) < (1, 0).$$

Indeed, agreement below $(1, 0)$ would put every e_n into the finite-dimensional span of the coefficients of y , a contradiction. Admitting the scale $t^{(1,0)}$ therefore defeats *all* such approximants, not just the original partial sums. The latter still have error valuations $(0, N + 1)$, all below $(1, 0)$.

In addition, Theorem 6.4 supplies a continuous K_Δ -linear functional L on V_Δ with

$$L|_V = 0, \quad L(x) = t^{(1,0)}.$$

Its restriction

$$L|_{V_\mathbb{Q}} : (V_\mathbb{Q}, \tau_\mathbb{Q}) \longrightarrow (K_\Delta, \tau_\Delta)$$

is not continuous, where each τ is the intrinsic valuation topology of its indicated group. Indeed, $x_N \rightarrow x$ in $\tau_\mathbb{Q}$ and $L(x_N) = 0$; continuity into the Hausdorff target would force $L(x) = 0$, contrary

to $L(x) = t^{(1,0)}$. The target is K_Δ , not $K_\mathbb{Q}$, since that value is outside the old field. Using only old rational thresholds on K_Δ would instead give a non-Hausdorff topology: $t^{(1,0)}$ would lie in every zero neighborhood. There is no contradiction with continuity on V_Δ : its induced topology on $V_\mathbb{Q}$ is discrete, as just proved.

The same two-scale group already appears in the collection for scalars: the notation guide notes that in $\mathbb{C}((t^{\mathbb{Q}+\mathbb{Q}\omega}))$ the finite partial sums of $\sum_{n \geq 0} t^n$ have error valuations that never exceed ω , and the spectral report proves that such geometric partial sums do not converge in the valuation topology although the family is strongly summable (`ihf:rf:prop:noconv`). The example above is the vector-valued form of that phenomenon with independent coefficients, where it removes x from the completion altogether.

10.3 What remains coherent under extension

The support expansion of a strong operator $A \in \text{Hom}_k(V, W)((t^\Gamma))$ is still a well-ordered expansion after embedding Γ into Δ . It therefore defines a strong K_Δ -linear operator on V_Δ , and its action on V_Γ agrees with the old action: every coefficient is computed by the same finite convolution. This extension is the unique *strong* K_Δ -linear extension, since its values on constants determine its operator coefficients by Theorem 3.2. Composition is coherent for the same finite-coefficient reason.

For a represented functional, use the embedded representer y : $L_y^\Delta(x) = \langle x, y \rangle_\Delta$ restricts to L_y^Γ . An ordinary algebraic functional likewise keeps its coefficient map, including when it is unbounded in the ordinary norm. For the hyperplane construction, a fixed choice of φ and u gives $M_\Delta \cap H_\Gamma = M_\Gamma$, by the coefficientwise kernel condition, and the formula $x - u\hat{\varphi}(x)$ shows that the enlarged projection restricts to the original one.

By contrast, an invisible continuous functional comes from extension choices on vectors not determined by constants. Its definition does not supply a canonical coefficientwise enlargement, and the completion on which it must vanish changes exactly as (10.2) specifies. We claim no canonical transport for such functionals.

11 Interpretation inside the surreal and surcomplex numbers

11.1 Fixed Hahn workspaces give genuine surreal scales

The link to surreals is the normal-form embedding. For a set-sized ordered additive subgroup $\Gamma \subset \mathbf{No}$, send

$$t^\gamma \longmapsto \omega^{-\gamma}. \quad (11.1)$$

A well-ordered support in Γ becomes a reverse-well-ordered set of Conway exponents. The normal-form theorem and its arithmetic therefore give an ordered-field embedding

$$\Phi_\Gamma : \mathbb{R}((t^\Gamma)) \longrightarrow \mathbf{No}, \quad \Phi_\Gamma \left(\sum_\gamma^H r_\gamma t^\gamma \right) = \sum_\gamma^H r_\gamma \omega^{-\gamma}.$$

Uniqueness of normal forms gives injectivity. Addition is coefficientwise; monomial multiplication and finite convolution preserve products. The first nonzero real coefficient determines the sign on both sides, so the embedding preserves order. Splitting complex coefficients into real and imaginary parts gives the embedding into $\mathbf{No}[i]$, preserving conjugation. Admissible strong families retain their support union and finite fibers, so these embeddings also preserve their strong sums. This uses normal-form arithmetic, not convergence of finite subsums in the full surreal fine topology.

The sign convention is exactly that of [6, Sections 2.2 and 2.4] and of the spectral report's `ihs:eq:surreal-embedding`. The repository's scalar normal-form layer implements this bridge; current implementation pointers are in `docs/NORMAL_FORM_BRIDGE.md`, while [7] retains the manuscript's repository pin. The bridge's availability does not formalize the vector-module results proved here. For this application one may take divisible groups, so that the scalar workspaces are also real closed and algebraically closed, respectively; the main duality proofs did not require that extra hypothesis.

The Hilbert coefficients themselves remain vectors in a fixed set-sized ordinary space H . We are constructing a module over an actual surcomplex subfield, not claiming that a Hilbert vector is a single surcomplex scalar. Inner products, norms and scalar-valued functional outputs lie in those subfields. General operator outputs remain module vectors; their norm inequalities compare actual surreal scalars.

11.2 The two-scale example in Conway notation

For the preceding example take

$$\Gamma = \mathbb{Q} \subset \mathbf{No}, \quad \Delta = \mathbb{Q}\omega + \mathbb{Q} \subset \mathbf{No},$$

with the natural lexicographic ordering under $(a, b) \mapsto a\omega + b$. Then (10.3) becomes

$$x = \sum_{n \geq 1}^H \omega^{-n} e_n. \quad (11.2)$$

It is a limit of finite sums for the intrinsic rational-exponent valuation topology. After admitting the exponent ω , no finite coefficient-span vector approximates it to valuation ω . The new scalar tolerance is $\omega^{-\omega}$, and an invisible functional in the enlarged workspace can satisfy $L(x) = \omega^{-\omega}$ while vanishing on every constant vector.

The support of (11.2) and its strong sum have not changed. What changed is the family of precision demands. This gives a concrete obstruction to constructing infinite-dimensional surcomplex analysis by completing finite-coefficient data at one stage and assuming the completion is unaffected by larger surreal scales.

11.3 Three scope restrictions

All theorems above concern set-sized supports, exponent groups, vector spaces, and index sets. They are ordinary ZFC statements. They do not apply Zorn's lemma to a proper class of vectors, and they do not build a continuous dual of a class-sized space over the whole of $\mathbf{No}[i]$.

The intrinsic topology on a fixed Hahn workspace must not be identified with the subspace topology inherited from the full surreal fine topology. Choose a surreal θ above every member of the set Γ . Every nonzero real or complex scalar in the workspace has leading valuation $\gamma \in \Gamma$, so its absolute value exceeds $\omega^{-\theta}$. The fine ball with that positive tolerance therefore meets the scalar workspace only at zero, and translation shows that the induced fine topology is discrete. For H_Γ , the corresponding statement uses *norm balls* with all positive surreal tolerances: $v(\|x\|) = v(x) \in \Gamma$ for $x \neq 0$ gives the same isolation. It does not embed Hilbert vectors as scalars. The repository already emphasizes the distinction between strong summability, workspace convergence, and full-surreal fine convergence [8]. Our completion-enlargement theorem is an algebraically precise instance of that distinction.

Finally, coarse order-boundedness is not an ordinary operator norm estimate with a real constant. A positive infinite bound hides all ordinary finite coefficient growth. Extra hypotheses are required to recover the ordinary analytic meaning of boundedness, adjoints, or Riesz representation. The coefficientwise criterion Theorem 8.4 states exactly one such hypothesis.

12 The report in the collection

12.1 Relation to infinite-dimensional Hahn spectral theory

The Hilbert results of sections 8 and 9 live in the space $H_\Gamma = H((t^\Gamma))$ of Part I of the spectral report [9], with the same inner product `ihh:hh:eq:inner`, the same linearity convention and the same field F of norm values. That report continues to receive additions, so the statements below cite it only by label, never by number. For a Hilbert coefficient space the three operator classes are nested:

$$\mathcal{B}(H)((t^\Gamma)) \subseteq \text{End}_\mathbb{C}(H)((t^\Gamma)) \subseteq \text{End}_K^{\text{cont}}(H_\Gamma).$$

The first is the algebra of everywhere-defined adjointable operators (`ihh:hh:thm:adjoint`); the second is the algebra of strong endomorphisms (Corollary 3.3); the second inclusion, that strong maps are continuous, is the last sentence of Corollary 3.3. That adjointable operators are strong is also the first sentence of `ihh:hh:cor:autobounded`. The first inclusion is proper when H is infinite-dimensional, since an unbounded algebraic endomorphism is strong but not adjointable, as `ihh:hh:rem:baire` anticipates. The second is proper when in addition Γ is noncyclic: if L is a continuous functional that is not strong (Theorem 7.3) and $u_0 \neq 0$ is a constant, then $x \mapsto L(x)u_0$ is continuous but not strong, because composing it with the coefficientwise extension of an algebraic functional taking the value 1 at u_0 , which is strong, returns L . The algebraic operators used in the spectral report’s range-defect theorem (the left inverse U , the maps R , L and the projection Π of `ihh:hh:thm:defect`) all lie in the middle algebra: they are strong, and in general not adjointable.

For Part II of the spectral report the relevant coefficient space is $V = \mathbb{C}^{(I)}$, with no Hilbert structure. Theorem 3.2 applies to it: the strong endomorphisms of $\mathbb{C}^{(I)}((t^\Gamma))$ are the Hahn series of arbitrary linear endomorphisms of $\mathbb{C}^{(I)}$, that is, of column-finite matrices. The algebra $\mathcal{A}_I$ of Hahn series of row- and column-finite matrices is a subalgebra, and the sufficiency half of Theorem 3.2 contains the operator-action part of `ihh:rf:lem:strongaction`. Sections 4 and 6 apply to $V = \mathbb{C}^{(I)}$ as to any vector space. Section 8 assumes a complete Hilbert coefficient space and is not asserted for $\mathbb{C}^{(I)}$ with infinite I .

The range-defect and spectral theorems of that report are not used and are not reproved here. Conversely, nothing here bears on its spectral formulas: a spectrum is a statement about bijectivity or invertibility, and this report makes no spectral statement.

12.2 Other neighbours

The report on entire functions at arbitrary rank uses the same two order-theoretic notions, the cofinality $\text{cf}(\Gamma)$ and cofinality of a subgroup, with the same meanings; the two exact statements that share an invariant are compared in Remarks 4.7 and 10.2. The dynamics report’s “strongly linear” maps (`dyn:def:summable`) are defined by the same two conditions; the operator classification here additionally requires linearity over the whole Hahn field K . The collection’s notation guide, `docs/NOTATION.md`, and its report inventory, `docs/README.md`, record the three topologies and the distinction between strong summation and convergence that section 11 relies on.

Spectral report	Here	Relation
<code>ihs:lem:support</code>	Lemma 2.2	Same classical lemma; here with the two-set proof
<code>ihs:hh:prop:inner</code>	Proposition 8.1	Same statement, repeated with proof
<code>ihs:warn:three</code>	Theorems 4.2 and 4.6	“ H_Γ is not $H \otimes_{\mathbb{C}} K$ ” made exact: $E_\Gamma(H) = K \otimes_{\mathbb{C}} H$, its closure, and when each equals H_Γ
<code>ihs:hh:thm:adjoint</code>	Theorem 3.2	Adjointable $\subseteq$ strong; Baire argument versus descending-support argument
<code>ihs:hh:prop:complete</code>	Proposition 5.1	Rank-one metric completeness versus spherical completeness at every rank
<code>ihs:hh:cor:autobounded,</code> <code>ihs:hh:thm:norm</code>	Proposition 8.2 and Theorem 8.3	Bounds for adjointable versus all continuous maps; least bound for bounded versus unbounded coefficients
<code>ihs:hh:prop:riesz</code>	Theorem 8.4, Remark 8.5	That functional is a nonzero element of the first gap $D_{\text{str}}/D_{\text{rep}}$
<code>ihs:hh:thm:closedrange</code>	Theorem 9.1	Closed subspaces with zero orthogonal complement; range of infinite codimension versus a hyperplane
<code>ihs:hh:cor:extension</code>	Theorem 10.1	Constant-normal spectra and the stated positive-order range defects persist; completions have a cofinal/noncofinal dichotomy
<code>ihs:rf:lem:strongaction,</code> <code>ihs:rf:prop:noconv</code>	Theorems 3.2 and 7.3	For $V = \mathbb{C}^{(I)}$: row- and column-finite Hahn matrices are strong; strong sums need not converge in valuation off the cyclic case

Table 1: Where each nearby statement of the spectral report sits relative to this report. These comparisons do not derive the present duality classification from the spectral theorems; shared classical support inputs are identified separately.

The later first- κ report [11] supplies a related cardinal-support criterion. For uncountable κ , $k \in \{\mathbb{R}, \mathbb{C}\}$ and nonzero divisible Γ , its completed field consists of series with fewer than κ support positions below every exponent (`fkf:thm:completion`). Its exact cofinality test for completeness assumes that the field of series with fewer than κ terms is *proper* in the full Hahn field (`fkf:thm:dichotomy`). If it already equals the full Hahn field, it is complete regardless of that test. The object and invariant differ from finite coefficient rank here; `fkf:rem:countable-analogue` explicitly makes this comparison. Neither result is used to prove the other.

13 Worked checks and a formalization route

13.1 What the accompanying program does

The accompanying `code/verify_examples.py` uses exact rational arithmetic and finite-dimensional rational linear algebra. It checks finite coefficient convolution, composition and scalar compatibility, leading exponents of squared norms, coefficient-rank tests, and finite instances of the projection and two-scale formulas. Its output is recorded in `data/verification_report.json`: 610 exact assertions in nine groups (operator composition, scalar compatibility and finite-family regrouping, 60 each; leading squared norm, 180; coefficient ranks below cuts, two-scale ranks and projection algebra, 69 each; a 40-term prefix of the descending-support family of Theorem 7.1(b), 42; one

cancellation check). The random inputs use the fixed seed 20260922. The report also lists what the program does not verify.

The program does not claim to certify the infinite theorems. Well-ordering arguments, spherical completeness, Hamel-basis extension, and the existence of invisible functionals are proved in the text. In particular, no finite truncation can witness an invisible functional that is zero on constants but nonzero on a Hahn sum: a finite truncation is a finite K -linear combination of constants and must be annihilated. This limitation is itself a consequence of linearity, not an implementation defect.

13.2 A small finite calculation that does carry information

Let $V = \mathbb{Q}^3$, $W = \mathbb{Q}^2$, and use two-dimensional rational lexicographic exponents. A finite operator series $A = A_0 + t^{(0,1)}A_1 + t^{(1,0)}A_2$ and finite vector series $x = x_0 + t^{(0,2)}x_1$ give

$$\begin{aligned} Ax = & A_0x_0 + t^{(0,1)}A_1x_0 + t^{(0,2)}A_0x_1 + t^{(0,3)}A_1x_1 \\ & + t^{(1,0)}A_2x_0 + t^{(1,2)}A_2x_1. \end{aligned}$$

The program draws operator and vector series with supports among the four exponents $(0,0)$, $(0,1)$, $(0,2)$, $(1,0)$ and random rational coefficients, and compares this coefficient-level action with the action of composed operator series. Cancellations are retained rather than assuming every possible support sum is occupied. Rank checks similarly use actual linear dependence, not just the number of nonzero coefficients. For example, arbitrarily many coefficients of the form $c_n u$ have rank one and do not obstruct algebraic scalar extension.

13.3 Dependencies for a Lean development

A feasible formalization separates three layers. First comes the purely algebraic Hahn-vector layer: well-ordered supports, finite fibers, coefficient maps, convolution, and the descending-support test in Theorem 3.2. This layer needs no Hilbert space. Second comes the valuation layer: balls as matching truncations, coefficient-rank closure, spherical completeness, and the one-step extension theorem followed by Zorn. Third comes the Hilbert layer: the coefficient form, its binomial norm, ordinary Riesz representation, and the exact dual comparison.

The repo already has substantial scalar Hahn infrastructure and an independent normal-form bridge [7]. Those are reusable prerequisites, not a claim that the new module theorems are already formalized. In the current tree the *Implementation mappings* section of `docs/FORMALIZATION.md` has no mapping for a label of this report. Its statement index does list the labels; an inventory entry is not a proof-status claim. The nearest formalized prerequisite is scalar: Hahn-to-Hahn embeddings along strictly increasing exponent maps preserve strong summability and Hahn sums (`Surreal/HahnSeries/WorkspaceEmbedding.lean`, recorded in that ledger), which is the ring-coefficient case of the embeddings used in section 10. A useful first milestone would be the finite coefficient-rank criterion and the cofinal/noncofinal intersection theorem: they isolate the key obstruction without first developing an entire operator theory. The nonconstructive steps should be explicit uses of classical choice, not hidden as executable definitions of arbitrary functionals.

14 Limits, further questions, and the audit

14.1 What the article settles and what remains separate

Within the specified Hahn-vector model, the completion question, strong-Hom question, image and kernel of continuous restriction, existence of invisible functionals, and scalar-enlargement question are settled by the displayed theorems. The Hilbert specialization determines exactly which strong functionals are represented and exactly when the three duals coincide.

Several separate problems remain worthwhile. One is to classify additional regularity conditions on restrictions to constants that recover a useful intermediate dual while remaining stable under noncofinal scalar enlargement. A second is to describe the size and possible canonical structures of $N_\Gamma(V)$ beyond its exact annihilator characterization. A third is to develop operator ideals in this coefficient model that retain ordinary compactness or nuclearity without mistaking valuation continuity for an ordinary norm estimate; the relevant caution is the spectral report’s, that “compact” in `ihs:hh:cor:compact` describes the ordinary operator and asserts nothing about compactness or compactoidness in the valuation topology. These are proposed continuations, not assertions of named published open problems or prerequisites for the results proved here.

14.2 Repository comparison

The repository was inspected at the fixed revision

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The principal comparison sources were the root README, the complete report inventory in `docs/README.md`, the introduction and relevant Hahn–Hilbert construction/operator sections of `docs/surcomplex/infinite-dimensional-hahn-spectral-theory/article.tex`, and the introductions to the differential-equations and analytic-geometry reports. The latter two were examined to avoid reusing directions already substantially developed in the project. The report inventory provides the comparison map [8].

The closest pre-existing result is the automatic adjoint theorem in the infinite-dimensional spectral report. It identifies adjointable endomorphisms with $\mathcal{B}(H)((t^\Gamma))$ and uses ordinary closed-graph and Baire arguments. The present strong-Hom theorem identifies *strong* endomorphisms with the larger algebra $\text{End}_{\mathbb{C}}(H)((t^\Gamma))$ by a different, purely support-theoretic argument. The principal additional results concern the completed constant span and the continuous dual beyond that algebra. We found no statement of the exact coefficient-rank completion theorem, the restriction exact sequence with its two failure mechanisms, or the cofinal/noncofinal completion-intersection formula in the inspected material. This is a targeted content comparison, not an exhaustive audit of every line or historical revision of the repository.

Update at placement. The paragraphs above describe the repository at the manuscript’s pin and are kept as provenance. When this report was placed (collection revision `52c7ab6`) the comparison was repeated against the current tree, with the following changes. (i) The spectral report’s revision `6e34651` removed Part I’s standing divisibility hypothesis and proved closedness of its coercive range at every rank (`ihs:hh:thm:closedrange`); at the pin that closedness was proved for divisible $\Gamma \subseteq \mathbb{R}$. (ii) The manuscript’s comparison does not mention two statements that were already present at the pin and are the closest in-collection predecessors of its Hilbert results: the explicit non-represented continuous functional `ihs:hh:prop:riesz`, which Remark 8.5 identifies as an element of

the first gap, and the closed subspace with zero orthogonal complement `ihs:hh:thm:closedrange`, compared with Theorem 9.1 after its proof. (iii) The collection’s report on entire functions at arbitrary rank uses the same cofinality invariants and proves a change-of-workspace formula of the same shape for entire power series (Remarks 4.7 and 10.2). (iv) At placement, two further manuscripts were added to the spectral report (on Drazin invertibility and on compact self-adjoint operators under noncommuting perturbations); a search of both manuscripts found no theorem about duals, completions of the constant span, or strong maps. Both use “Riesz” for finite Riesz spectral projections and decompositions; in this report “Riesz” always means Riesz representation of functionals. A search of the tree at that placement found no other statement of the coefficient-rank completion theorem, the restriction exact sequence, or the completion-intersection formula. This is a historical finding with the same targeted-comparison caveat, not a renewed claim about every later revision. In particular, the subsequently assembled first- κ report now summarizes the completion results here and proves a distinct cardinal-support analogue, as described in section 12.

Topic	Established input or nearby result	Role in this article
Strong summability	Hahn theory; BKKPS summability spaces	Definitions and support machinery, not new
Hahn–Banach extension	Ingleton; Morillon’s choice discussion	Full needed proof given; classical mechanism
Hahn–Hilbert adjoints	Repository spectral article (<code>ihs:hh:thm:adjoint</code>)	Distinguished from strong and continuous maps
Completeness	Repository spectral article, rank one (<code>ihs:hh:prop:complete</code>); classical spherical completeness	Spherical completeness at every rank, with proof
Completion	Algebraic scalar extension inside a Hahn module	Exact finite-rank-below-cuts criterion and cyclic boundary
Continuous restriction	Uniform valuation estimates and extension	Exact image, invisible kernel, two independent obstructions
Exponent enlargement	Coefficient-preserving Hahn embeddings; the entire-functions change-of-workspace criterion	Cofinal preservation versus noncofinal collapse
Riesz and orthogonality	Ordinary Riesz; known non-Archimedean failures; <code>ihs:hh:prop:riesz</code> and <code>ihs:hh:thm:closedrange</code> in the same space	Exact quotient and explicit hyperplane/distance-cut package
Field-valued bounds	<code>ihs:hh:cor:autobounded</code> , <code>ihs:hh:thm:norm</code> for bounded coefficients	Bounds for all continuous maps; the unbounded-coefficient case

Table 2: Attribution and the proposed contribution. Closely related classical tools are not recast as new theorems.

14.3 Literature and verification boundary

The manuscript’s literature check consulted the primary summability article [1], the surreal automorphism article [6], Morillon’s paper [3], and publication records for the neighboring non-Archimedean Hilbert literature [4]. Bibliographic details of Ingleton’s original paper are corroborated in Morillon’s references; we did not obtain its full text. Some publisher pages exposed only metadata. Search results that were irrelevant or inaccessible were not treated as evidence of absence. The references

below identify the actual theoretical predecessors rather than trying to certify priority by a keyword count.

The subsequent review checked [1, Sections 1.1–1.3], including Definition 1.19’s two requirements for strong linearity and Example 1.7’s Hahn support conditions. That paper refers to earlier nonstrong-map examples rather than presenting one there. It also checked [6, Sections 2.2 and 2.4] for the normal-form sign convention and [3, Section 3.1.2, Lemma 2] for the real-valued one-step extension statement. The ordered-group proof here supplies its own ball construction; the cited real-valued formulation is not silently applied at arbitrary rank. These targeted primary-text checks do not certify priority or complete the audit of all imported results. The delivered `source_audit.md` remains unchanged as historical evidence.

The main arguments have been checked internally for their hypotheses, including arbitrary rank, noncofinal inclusions, zero maps, finite coefficient dimension, and the cyclic exceptional case. The exact finite examples have been executed. Neither activity is independent refereeing, and no Lean verification of these new statements is claimed. The package is a research manuscript with complete proofs, not a certified resolution of a longstanding named conjecture. When the report was placed in the collection, the proofs were read again against their statements, and the finite program was rerun; its output agrees with the recorded report. No mathematical correction was needed. That reading is an editorial check, not refereeing.

15 Conclusion

There are two independent reasons why extending a Hilbert space to Hahn scalars does not transport the classical duality picture unchanged. First, valuation boundedness ignores ordinary coefficient growth: an ordinary discontinuous algebraic functional has a strong coefficientwise extension bounded by every positive infinite scalar. Second, valuation completion does not exhaust strong Hahn summation: a vector with infinitely many independent coefficients below one cut cannot be reconstructed from finite coefficient spans, and continuous functionals can detect precisely this defect while vanishing on every constant.

The organizing formulas are

$$\mathrm{Hom}_K^{\mathrm{str}}(V_\Gamma, W_\Gamma) = \mathrm{Hom}_k(V, W)((t^\Gamma))$$

under convolution and

$$C_\Gamma(V) = \{x : \text{the coefficient rank below every cut is finite}\}.$$

Their interaction yields the exact continuous restriction sequence, the three-dual trichotomy, and a sharp failure of completion under noncofinal enlargement. The concrete tolerance $\omega^{-\omega}$ in equation (11.2) illustrates why these distinctions matter specifically at surreal scales: adding a new scale can invalidate an old completion without changing a single coefficient of its strong sum.

A Provenance, editorial changes, and non-claims

A.1 The source

This report is built from one manuscript, *Three Duals at Surreal Scales: Strong Hahn Operators, Completion Defects, and Invisible Functionals*, dated September 22, 2026 and prepared for Vladimir Reshetnikov. It arrived as item 04 of the collection’s nineteenth batch of manuscripts and was written against repository revision 048b72cf7cbfc8ab246e4f73788c10460cb3f6e0. It was placed in the collection at revision 52c7ab6d3bbf3a9c5e6142da055947acc9fedff5. It contributes the entire mathematical content: every definition, theorem, proof and example of sections 1 to 11, 13 and 14, the program `code/verify_examples.py`, its recorded output `data/verification_report.json`, and the audit file `source_audit.md`. These three files are kept exactly as delivered. The manuscript’s own PDF and member README are not shipped; the README of this directory is new, and the PDF is rebuilt from this source. There was no second source, so nothing had to be printed once from two sources, and there is no second proof of any result.

A.2 Editorial changes

1. All 70 delivered labels now carry the prefix `duals::`; none was otherwise renamed or removed. New labels: `duals:sec:conventions`, `duals:rem:order-unit`, `duals:rem:ihh-riesz`, `duals:rem:basechange-collection`, `duals:sec:collection`, `duals:tab:ihh` and `duals:app:provenance`.
2. Notation: the Hahn–Hilbert space is written H_Γ throughout (the manuscript used both H_Γ and $\mathcal{H}_\Gamma$); the hyperplane projection of Theorem 9.1 is written P_M , because $P_\Gamma(V)$ denotes the restriction space.
3. Terminology is fixed once in section 1.4. The sufficient condition in Theorem 4.5 is identified with the order unit of the entire-functions report, and Remark 4.7 explains why countable cofinality is the exact condition.
4. The repository comparison of section 14 is kept as provenance and followed by an update against the current tree. Pointers to the spectral report were added at Lemma 2.2, Propositions 5.1, 8.1 and 8.2, and Theorems 8.4, 9.1 and 10.1 and in sections 11 and 13; section 12 and table 1 are new. Remark 8.5 is an editorial comparison: it shows, from the two cited statements, that the spectral report’s non-represented functional is strong. It adds no hypothesis and changes no theorem.
5. The checks section names the file locations and the per-group assertion counts from the recorded output.
6. The title page now records the draft status and the provenance, and the abstract adds the countable-cofinality clause, which is Theorem 4.6, and a pointer to section 12.

At placement, no statement of a theorem, proposition, lemma or corollary was strengthened or weakened, and no proof was changed. Where a choice had to be made, the delivered title was kept, the proof of Proposition 8.1 was kept even though the same statement is `ihh:hh:prop:inner`, and the delivered audit file keeps its own pin.

The subsequent proof review covers the main text, Sections 1–15. It expands support and regrouping, uniform shifts, global operator support, coefficient-rank completion, cofinality, compatible

truncations and the set-sized Zorn argument, including zero cases. The order-unit summary distinguishes cyclic from noncyclic groups; the choice explanation is corrected. The duality proofs give restriction bounds, coefficient-rank separation, the canonical quotient map and cyclic finite-subsum nets.

The Hilbert proofs expand norm identities, neighborhood comparisons, coefficientwise Riesz and cardinality. The spectral norm comparison now uses nonnegative bounds, including zero; bounds for an unbounded leading coefficient are characterized by their valuations. The hyperplane proof gives its algebraic direct sum, distinguishes orthogonal complements from orthogonal direct-sum complements, and computes distance valuations. The distance theorem repeats the existing nonzero-group convention. The enlargement proof identifies both induced topologies, the target of the two-scale restriction, and the unique strong operator extension. The normal-form discussion distinguishes scalar values from module vectors and proves isolation by full-surreal tolerances. Source comparisons now state the first- κ hypotheses, pin historical absence claims, and separate implemented scalar prerequisites from new module results. All labels, result numbers and delivered audit/verification artifacts are preserved. The targeted literature checks above do not complete the remaining imported-result and source-reconciliation work.

A.3 Non-claims

Every limitation stated by the manuscript is kept at its point of use. Collected here, they are the following.

1. The contribution is the combined completion–duality–enlargement classification; Hahn summation, non-Archimedean Hahn–Banach and failures of Riesz representation are not claimed as inventions (title page, section 1).
2. No named published conjecture or open problem is claimed solved.
3. The repository and literature comparisons are targeted; they do not certify originality or priority, and an equivalent statement may exist in print.
4. The proofs have not been independently refereed and have not been verified in Lean or any other proof assistant; no label of this report has a Lean mapping.
5. The repository’s automatic-adjoint and Hahn–Hilbert results, and the other spectral-report statements compared in table 1, are not claimed as new.
6. Formal summability spaces, strongly linear maps and their operator algebras are established notions [1].
7. Spherical completeness and extension are tools, and non-Archimedean failures of orthogonal complementation are established [4].
8. The positive-monoid support lemma is classical and used only for square roots (Lemma 2.2).
9. The Hahn–Banach extension uses Zorn’s lemma; no effective procedure is asserted, and invisible functionals are existence results, not computable formulas (section 5).
10. The exact restriction sequence has no specified canonical splitting and no preferred extension operation (Theorem 6.3).
11. In Theorem 7.1(a) summability is preserved only for the displayed family; logical independence of the two clauses of the definition of strongness is not asserted.

12. No real-Hahn operator norm is defined as a least bound; the set of bounds need not have a least element (Proposition 8.2).
13. No ordinary continuity of the Hamel-basis functional of Theorem 8.3 is asserted.
14. No equality is asserted in the cardinal bound of Proposition 8.7 for arbitrary Γ .
15. Theorem 10.1 concerns intersections of embedded sets; no commutation of a completed tensor-product functor with enlargement is claimed.
16. No canonical transport of invisible functionals to a larger workspace is claimed (section 10).
17. A Hilbert vector is not claimed to be a surcomplex scalar; the construction is a module over a surcomplex subfield, through fixed normal-form workspaces (section 11).
18. No Zorn argument over a proper class and no continuous dual of a class-sized space over $\mathbf{No}[i]$ is constructed.
19. The intrinsic topology of a workspace is not the subspace topology of the full surreal fine topology.
20. Order-boundedness is not an ordinary operator-norm estimate; the ordinary meaning of boundedness, adjoints or Riesz representation needs extra hypotheses.
21. “Invisible on constants” is a definition, not a statement that the functionals are zero; the constants need not be topologically total in the Hahn space.
22. The Hahn field does not replace the ordinary zero distance of Theorem 9.2 by a particular infinitesimal.
23. The program checks finite identities and finite prefixes only. It does not verify infinite well-ordering or the arbitrary-rank Hom theorem, the completion and cofinality classifications, spherical completeness or Hahn–Banach extension, the existence of Hamel or invisible functionals, the orthogonal-complement and missing-infimum claims, or novelty. Its finite projection check claims no finite analogue of the zero orthogonal complement.
24. The formalization route of section 13 is a proposal; existing Lean infrastructure is a prerequisite, not a formalization of these theorems.
25. The further questions of section 14 are proposed continuations, not named open problems and not prerequisites.
26. The full texts of Ingleton’s paper and of [4] were not obtained, Neumann’s paper was not audited in full, and inaccessible pages or empty searches were not treated as evidence of absence.

The editorial additions carry their own limits: the comparisons of section 12 and Remarks 4.7 and 10.2 are comparisons of statements, not reductions of one theorem to another.

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